

**"STUDIES ON DRY ROOT ROT OF CHICKPEA INCITED BY
Rhizoctonia bataticola (Taub) Butler."**

DISSERTATION

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Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani
In partial fulfillment of the requirements
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**MASTER OF SCIENCE
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IN

PLANT PATHOLOGY

By

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2016**

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*I hereby declare that the dissertation
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This is to certify that the dissertation entitled “STUDIES ON DRY ROOT ROT OF CHICKPEA INCITED BY *Rhizoctonia bataticola* (Taub) Butler” submitted by Miss. **KADAM AMRUTA MADANRAO (21Mb/2014)** to the College of Agriculture, Badnapur in partial fulfillment of the requirement for the degree of **MASTER OF SCIENCE (Agriculture)** in the subject of **PLANT PATHOLOGY** has been approved by the student's advisory committee after viva-voce examination in collaboration with the external examiner.


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(Kadam A.M.)

ABBREVIATIONS

%	- Per cent
/	- Per
C.D.	- Critical differences
Cm	- Centimeter
d. f.	- Degree of freedom
e.g.	- Exempli gratia (For example)
et al.	- Et alia (and others)
etc.	- Et cetra
Fig.	- Figure
Ha	- Hectare
i.e.	- That is
J.	- Journal
Kg	- Kilogram
M	- Million
M / ha	- Million per hectare
m ²	- Square meter
MS	- Mean square
MSe	- Error mean squares
MT	- Metric Tonnes
q/ha	- Quintals per hectare
Sci.	- Science
SE	- Standard Error
SE(d)	- Standard error of difference
SE(m) ±	- Stanadard error of mean
Sr. No.	- Serial number
Unpub.	- Unpublished
Viz.,	- Videlicet (namely)
Vs	- Versus
Σ	- Standard deviation
Σ	- Summation

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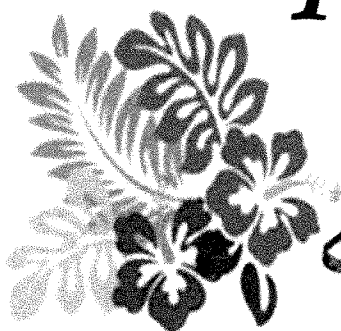
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Introduction



CHAPTER-I

INTRODUCTION

The cultivated chickpea (*Cicer arietinum* L.) was one of the first grain legumes to be domesticated in the old world. Chickpea is most probably originated in area of present day south eastern Turkey and adjoin Syeria.

The genus *Cicer* belongs to family Leguminaceae and sub family Papilionoidae. Chickpea has been well recognized as a valuable source of protein particularly in the developing countries , where majority of the populations depend on the low priced food for meeting their dietary requirements nutritionally. Chickpea is low in sodium (21.1%), fat (4.5%), contents has no cholesterol and overall an excellent source of both soluble and insoluble fiber , complex carbohydrates(61.5%), vitamins (especially B vitamins) and minerals (especially potassium , phosphorous, calcium, magnesium, copper , iron and zinc). Therefore, chickpea is an excellent heart healthy food that may be beneficial to the prevention of coronary and cardiovascular diseases and by reducing blood lipids also helps in some serious complications of diabetes.

Chickpea is grown in many tropical, sub-tropical and temperate regions of the world . Chickpea is mainly used for human consumption as well as animal feed. It is consumed as whole seed, dal, fried, boiled, salted or more generally cooked fresh green leaves generally used as vegetable. The grains also used as vegetable chole .Gram flour is mixed with wheat flour to improve the protein content of wheat flour and it is used for making missi roti . The flour of dehusked gram is called basen which is widely used in making pakodas, kadi, and several snack foods.

In India chickpea is primarily grown as *Rabi* (post rainy) season crop on residual soil moisture. It ensures nutritional security besides being a rich source of protein and is also important in substantial agriculture as it improves physical, chemical and biological properties of soil by mixing atmospheric nitrogen symbiotically. Deep roots of pulse crop also open up the soil by increasing soil aeration and fit well in various cropping ecosystem and hence have got unique position in rainfed agriculture.

Chickpea is grown throughout the world. In India it is grown on large scale in Punjab, Hariyana, Uttar Pradesh, Madhya Pradesh, Rajastan, Andhra Pradesh, Karnataka , and Maharashtra. It is mostly cultivated under rainfed condition in a variety

of soil, varying in a residual moisture. In India area under chickpea was 10.22 M/ha with production of 9.88 MT and productivity of 967 kg/ha during 2014. In Maharashtra chickpea is grown in *Rabi* season and it occupied area of 18.19 lakh ha with production of 16.22 lakh ton and productivity of 891 kg /ha during 2014. In Marathwada region, total area under chickpea during 2014 was 6.58 lakh ha with production of 6.64 lakh ton and productivity of 918 kg/ha. (Anonymous 2014)

However, chickpea productivity is low due to the susceptibility of the crop to different biotic and abiotic stresses. Abiotic stress is basically due to insufficient moisture. Regarding biotic stresses, diseases, insect pests, nematodes and parasitic weeds account major losses for example, extent of yield loss due to wilt and root rot disease is far more in the event of drought and high temperature in the country. The chickpea crop is attacked by 172 pathogens viz; 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma from all over the world (Nene *et al.*, 1996).

One of the major constraints limiting chickpea production is difficulty in managing diseases caused by pathogens especially soil borne. Important soil borne diseases of chick pea are - *Fusarium wilt* (*Fusarium oxysporium* Schlecht. Emnd synd. and Hans.f.sp.ciceri (Padwick) Synd and Hans), *Verticillium wilt* (*Verticillium albo-atrum* Reinke and Berth), dry root rot (*Rhizoctonia bataticola* (Taub) Butl: *Macrophomina phaseolina* (Tassi) (Goid), wet root rot (*Rhizovtonia solani* kulhn), black root rot (*Fusarium solani* (Malt) Apple and Wr,) *Phytophthora* root rot (*Pythium ultimum* Trow), collar rot (*Sclerotium rolfsii* Sacc.) and Foot rot (*Operculella padwickii* Kheswalla). These pathogens cause significant losses in yield and are primarily responsible for wide gap in yield levels in the farmers fields.

Chickpea dry root rot causing pathogen (*Rhizoctonia bataticola*) belongs to group of ubiquitous and diverse plant pathogens that occur widely in India as root pathogens on different crops and also isolated from a different varieties of same host species. Natural population of this pathogen is diverse in terms of pathogenic variability and can be characterized in to pathotypes or races by means of biological typing.

Being soil borne in nature it is very difficult to control this pathogen. Use of resistant cultivar is more effective strategy for sustainable chickpea cultivation. Despite of several resistant varieties are available, yield losses due to root rot are more. Ahmed

and Mohammed (1986) reported losses due to dry root rot in gram to the extent of 70.8% at full podding and 48.9% at pre-harvest stage. It indicates that use of resistant varieties is not a sole control measure against dry root rot of chickpea. Therefore, two or more control measures must be tried in combination with resistant variety for sustainable production.

Considering the behaviour of the pathogen, losses incurred in the farmers' field and limitations of individual control measures, the present investigation was planned with following objectives.

Objectives:

1. To survey chickpea fields for recording dry root rot incidence in Marathwada region.
2. To isolate and study the cultural and morphological characterization of *Rhizoctonia bataticola*.
3. To evaluate *in vitro* different fungicides, bioagents and organic amendments on growth of *Rhizoctonia bataticola*.
4. To evaluate effective fungicides, bioagents and organic amendments for integrated management of *Rhizoctonia bataticola* (pot culture).
5. To screen the different varieties/ germplasm lines against *Rhizoctonia bataticola* (pot culture).



*Review of
Literature*



CHAPTER-II

REVIEW OF LITERATURE

Chickpea (*Cicer arietinum* L.) is one of the important pulse crop grown in India and it is largely grown as rainfed as well as irrigated crop in *Rabi* season. Production of chickpea is affected by many soil borne pathogens. An attempt has been made in this chapter to compile all the relevant literature pertaining to dry root rot of chickpea caused by *R. bataticola* and its management under following sub-headings.

1. Occurrence
2. Yield losses
3. Survey and distribution
4. Isolation and pathogenicity
5. Symptomatology
6. Disease management
7. Screening of chickpea varieties, germplasm lines and genotypes

2.1 Occurrence

Agrawal and Sarbhoy (1976) reported that widespread occurrence of *Rhizoctonia bataticola* (*M. phaseolina*), causing charcoal rot of soybean and observed that the pathogen infects the crop in almost all stages of crop growth.

Saksena (1979) reported that *Rhizoctonia bataticola* (*M. phaseolina*) was the wide spread and destructive plant pathogen. The pathogen has been considered of great economic importance because, the pathogen was reported to cause root rot, collar rot, stalk rot, leaf spot and blight.

Vishwadhar and Sarbhoy (1989) reported that *Rhizoctonia bataticola* (Taub) Butler (Pycnidial stage, *M. phaseolina*) caused seedling rot, root rot, ashey stem blight and charcoal rot in the crops like soybean, mung bean, sunflower, urdbean etc. in India.

Srivastava *et al.* (2002) conducted survey in the Vindhyan plateaux zone (Bhopal, Raisen, Rajgarh, Sagar, Sehore and Vidisha) of Madhya Pradesh, India,

during 1998/99 to evaluate the prevalence of wilt and dry root diseases in chickpea. Some 50 plants showing wilt symptoms were collected from 23 locations. The greatest incidence of wilt was recorded from Sehore (25 per cent), followed by Rajgarh, Bhopal and Vidisha (20 per cent). Five fungi (*Fusarium oxysporum*, *Rhizoctonia bataticola* [*Macrophomina phaseolina*], *Trichoderma* sp., *Aspergillus* sp. and *Curvularia* sp.) were associated with chickpea samples. *F. oxysporum* was the most prevalent (56 per cent) in wilted plants, followed by *R. bataticola* (17 per cent) and *Trichoderma* sp. (10 per cent). The highest incidence of *F. oxysporum* was detected in samples from Bhopal and Sehore (67 per cent), followed by Raisen (60 per cent), Vidisha and Sagar (50 per cent), and Rajgarh (40 per cent). The incidence of *R. bataticola*, which also caused withering and drying of plants, ranged from 8 to 20 per cent. The occurrence of *Trichoderma* was higher in samples with lower incidence of *F. oxysporum* and *vice versa*.

Singh and Sirohi (2003) observed that dry root rot caused by *R. bataticola* was a serious disease in non irrigated chickpea areas. The incidence of disease was more in Una and Simour districts of Himachal Pradesh.

Gurha and Trivedi (2008) reported that *F. oxysporum* f.sp. *ciceri* and *R. bataticola* caused wilting and dry root rot diseases respectively in all the districts of Himachal Pradesh.

Rhiaz *et al.* (2011) conducted survey in Jammu and Kashmir. Dry root rot disease caused by *Rhizoctonia bataticola* was observed on variety Radhey in Jammu Kashmir.

2.2 Yield losses

Quantitative and qualitative yield losses incurred by root rot disease in chickpea, groundnut, black gram, green gram, cluster bean, sunflower; sesamum etc. were reported earlier by several workers.

Nene (1985) reported that chickpea root rot, collar rot and wilt were the most serious diseases which caused severe yield losses of 30-60 per cent under favorable condition.

Sen (2000) reported that *Macrophomina phaseolina* affected a variety of field and vegetable crops besides a wide range of urdbean plant species and caused charcoal rot, which accounted for crop losses to the tune of 10-20 per cent.

2.3 Survey and Distribution

Sharma *et al.* (1983) conducted survey of wilt and root rot incidence of *Cicer arietinum* in northern region of Madhya Pradesh. Survey was conducted in chickpea growing areas of 30 villages of northern region of Madhya Pradesh. The average of wilt incidence ranged from 14.90 to 37.99 per cent and root rot from 3.58 - 20.36 per cent.

Pande and Singh (1990) conducted survey in Allahabad region, India and observed chickpea infected by *F.oxysporum* and *R. bataticola*.

Srivastava *et al.* (2002) conducted survey in Vidhyan plateau zone of Madhya Pradesh, during 1998-99 to evaluate the prevalence of wilt and dry root rot diseases in chickpea. The incidence of *R. bataticola*, which also caused withering and drying of plants, ranged from 8 to 20 per cent.

Prajapati *et al.* (2003) conducted survey on incidence of chickpea dry root rot in Uttar Pradesh and Uttaranchal and reported the incidence of dry root rot of chickpea caused by *Macrophomina phaseolina* at 20 locations in Uttar Pradesh and Uttaranchal.

Prajapati *et al.* (2004) conducted survey on incidence of chickpea dry root rot in Uttar Pradesh. Survey was conducted at 20 locations of different agro-climatic regions of Uttar Pradesh. The disease was wide spread in occurrence in UP with 19.0 to 42.0 per cent incidence.

Kohire *et al.* (2006) carried out a survey in 2004-2006. They reported that wilt, dry root rot and stunt were associated with infected chickpea in Marathawada region of Maharashtra. Disease incidence varied from 1.6 to 16.5 per cent. Maximum average incidence was observed in Parbhani i.e. 18.5 per cent.

Rana *et al.* (2009) conducted survey on pea diseases and identification of fungi associated with wilt and root rot complex in Himachal Pradesh. The disease were prevalent in all pea growing areas of state during 2006-2008. The incidence of

wilt and root rot was 20 per cent in temperate wet, 15 per cent in sub temperate, 12 per cent in temperate dry and 11 per cent in subtropical regions.

2.4 Isolation and Pathogenicity

Bhate (1983) isolated *Rhizoctonia bataticola* from roots of green gram and proved its pathogenicity.

Sobti and Bansal (1988) isolated root, leaf and seed isolates of *R. bataticola* of groundnut and reported variation in cultural characters on different media.

Bansal *et al.* (1990) reported cultural variability among six isolates of *R. bataticola* isolated from different crops.

Devi and Singh (1998) studied on virulence of *Macrophomina phaseolina* isolates from black gram and green gram and reported that ,fifty six *Macrophomina phaseolina* isolates were obtained from roots, stems, pods and seeds of black gram and green gram from 11 different localities of North, South, and Central India. All isolates were categorized into different groups.

Katariya *et al.* (2007) isolated nine isolates of *R. bataticola* and tested pathogenicity of these isolates both in sterilized and unsterilized soil in chickpea variety C-325.

2.5 Symptomatology

Nene *et al.* (1978) found continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristic symptoms of gram infected by *R. bataticola*.

Casas and Diaz (1985) described symptoms caused by *Macrophomina phaseolina* as dry collar and root rots. Affected plants developed dark brown to black roots and dried up without showing any flaccidity.

Haware *et al.* (1990) reported that the disease development was influenced by dry soil condition especially at flowering stage. The plant suddenly dries in the field. The leaves and stem of affected plants usually become straw

colored. The tap root is dark and quite brittle in dry soil. The dark black sclerotia can be observed on and within host tissue.

Beniwal *et al.* (1992) reported that sudden drying of the whole plant was the most conspicuous symptom of this disease. The tap root of infected plant became dark showed signs of rotting, and was devoid of most of the lateral and finer roots. The dead root was brittle and its tip was easily broken.

2.6. Morphological and cultural characters *R. bataticola*

2.6.1. Morphological characters

Short *et al.* (1978) found that *M. phaseolina* pycnidiospores are ellipsoid to obovoid. During the sclerotial formation, 50–200 individual hyphal cells aggregate to give multicellular bodies named microsclerotia. The microsclerotia are black and are variable in size (50–150 μm) depending on the available nutrients of the substrate on which the propagules are produced.

Holliday (1981) described that *Macrophomina phaseolina* is well characterised by the presence of numerous black microsclerotia varying from 100 μm to 1 mm in stems, leaves and roots and 50 μm to 300 μm in culture. Under the microscope they appeared as round to ovate black dots which, when lighted from above, had relatively large, smooth, rounded bumps on their surface, a bit, like an artichoke.

Gupta and Kolte (1982) stated that *Macrophomina phaseolina* isolates produced abundant vegetative growth on PDA. It did not form sclerotia proportional to mycelial growth while the reverse way was true in the case of root isolates. The size of the sclerotia of the leaf isolates (98.10 μm) was more than that of root isolates (88.28 μm). They further reported that the two isolates differed from each other by their pycnidial size.

Mihail (1992) described that mature sclerotia of *Macrophomina phaseolina* were jet black and did not transmit light. Pycnidia might also sometimes be seen. These are black and globose varying from 100 μm to 250 μm in length with a truncate ostiole the spores of which are pointed at one end. The hyphae are septate, usually containing numerous vesicles.

Gaetán *et al.* (2006) reported that pycnidia of *M. phaseolina* are not produced on PDA except under some specific incubation conditions and only sometimes in host crops.

Sundravadana *et al.* (2011) described that *R. bataticola* seed isolate of blackgram and root isolate of soybean also recorded moderate sclerotial formation, while the shoot isolate of soybean produced less sclerotia. Among the isolates, the diameter of sclerotia ranged from 43.6 μm to 89.9 μm . The sclerotia of blackgram root isolate recorded a maximum diameter of 89.9 μm and were significantly superior and were followed by redgram root isolate (86.85 μm). The sclerotia of the shoot and root isolates of soybean were the smallest as 43.6 and 43.8 μm in diameter, respectively and were on par with one another. The shape of the sclerotia was categorized into two groups viz., round and irregularly oblong. The blackgram and redgram shoot recorded round sclerotia, while all the other isolates recorded irregularly oblong sclerotia.

Garg and Ashwini kumar (2012) reported that *Macrophomina phaseolina* differed widely in morphological and cultural characteristics. The fungal mycelium was hyaline and light to dark brown, profusely branched and constricted at septa. Sclerotia were oval to spherical, dark brown to black and measured 62.5 μm to 100 μm in diameter in culture.

Sundravadana *et al.* (2012) mentioned that ubiquitous, omnipresent sclerotial fungus *Rhizoctonia bataticola* (Taub.) Butl. in its pycnidial state is known as *Macrophomina phaseolina* (Tassi.) Goid. Isolates from different host species differ in their morphological and cultural characters and even differences occur in the isolates from various parts of same host.

2.6.2 Cultural charecteristics

EI- Wakil *et al.* (1985) concluded that PDA containing dextrose probably supports the growth of the pathogen. However, other media viz., Czapek's dox agar, Richard's medium are known to support good growth of *M. phaseolina*.

Mihail and Taylor (1995) reported that in PDA, pycnidia of *Macrophomina phaseolina* are not produced except under some specific incubation conditions and only sometimes in host crops.

Devi and Singh (1998) observed that good growth of *Macrophomina phaseolina* in PDA may be owing to provision of some additional nutrients in such media.

Lokesha and Benagi (2004) stated that differences were observed in radial growth of the *M. Phaseolina* isolates. Among the various solid media, the radial growth of isolates of *M. phaseolina* was maximum on PDA and Czapek's agar with mean colony diameter of 88.4 mm, which were significantly superior to all the other media, followed by Richard's agar (83.7 mm) and least growth of 63.5 mm was observed in Sabouraud's agar.

Sharma *et al.* (2004) reported that PDA was the best medium for growth and sclerotial formation of *R. bataticola*.

Salunkhe *et al.* (2009) found that out of six media tested, PDA supported the good growth and sclerotial formation of *Rhizoctonia bataticola* followed by Czapek's dox, Richard's agar, Oat meal agar and Peptone agar.

Khan *et al.* (2012) observed that potato dextrose agar was significantly superior than other media and supported the best growth of *Rhizoctonia bataticola* followed by nutrient agar and chickpea root extract agar medium. Fair growth of the pathogen was obtained on Asthana and Hawaker's media, whereas remaining media *viz.*, carrot root extract agar, oat meal agar, Czapek's Dox agar and Richard's agar supported poor growth. However, no sporulation was observed on any of the solid media.

Sundravadana *et al.* (2012) noticed that PDA medium recorded maximum mycelial growth of redgram shoot isolate and blackgram seed isolate of *R. bataticola* followed by Czapek-dox and Oat meal agar.

2.7 Disease management strategies

2.7.1 Evaluation of fungicides

Grewal and Vir (1958) found that Agrosan SW at 0.6 per cent completely inhibited *M. phaseolina* from growing out of seeds on potato dextrose agar medium.

Goyal and Mehrotra (1981) tested 9 fungicides against *R. bataticola* in *in vitro* for chickpea. The fungicides benomyl, carbendazim, thiophanate methyl effectively inhibited fungal growth.

Singh and chohan (1981) reported that Benlate at the concentration of 2.5 ppm, 5.0 ppm, 10.0 ppm and 20.0 ppm inhibited mycelial growth of *M. phaseolina* up to 85.0, 95.0, 97.8 and 100.0 per cent, respectively.

Taneja and Grover (1982) evaluated five formulations of four benzimidazole derived fungicides. Carbendazim, benomyl and thiophanate methyl were compared for their toxicity towards two pathogenic isolates of *Rhizoctonia solani* and three of *R. bataticola*. The isolates of two fungi showed significant differences in mycelial growth inhibition by the five fungicides. Benomyl and carbendazim were most inhibitory to all the isolates of both the fungi while the sesame isolate of *R. bataticola* was least sensitive to all the fungicides.

Sarwar and Gopala Raju (1985) studied the efficacy of various fungicides at different concentration against *R. bataticola* and showed that out of nine fungicides tested, Topsin M-70 was highly effective at 1000 ppm followed by formaldehyde, Aureofungin and Benlate at 3000 ppm. Brassicol was effective at 7000 ppm while rest of the fungicides proved to be ineffective even at higher concentration of 20000 ppm.

Ramados and Sivaprakasam (1987) tested carbendazim, quintozone and TMTD (thiram) at 250, 500 and 1000 ppm, and two granular insecticides, carbofuran and aldicarb at 250, 500, 1000 and 2000 ppm for *in-vitro* activity against *M. phaseolina*, a major pathogen of cowpea in TamilNadu. All the fungicides were inhibitory to linear growth of the pathogen. Carbendazim (1000 ppm) and thiram (500 ppm) were fungicidal while quintozone and the insecticides were fungistatic.

Bansal and Asha (1988) reported that linear growth of fungus was minimum (5mm) on benomyl and Dithane M-45 at 100 ppm. Brassicol and captafol gave good results at 50 ppm concentration whereas Dithane M-45, Captaf and Calirus did not show any inhibition of mycelial growth at this concentration.

Kakade and Khune (1989) tested 13 different fungicides against 7 phytoathogenic fungi *in-vitro*, of them delsan 30 EC was the most effective and inhibited all the test fungi at 0.25 and 0.30 per cent concentrations. Fosetyl aluminium

inhibited *Alternaria alternata*, *Curvularia lunata*, *F. moniliforme*, *R. bataticola* (*M. phaseolina*) and *S.rolfsii* at 0.2 and 0.3 per cent. Bavistin inhibited *A.niger*, *G. fujjikoni* and *M. phaseolina* completely.

Peshney *et al.* (1992) conducted *in vitro* studies with *R. bataticola* using fungicides thiram, captan, PCNB, mancozeb, iprodione, carboxin, carbendazim, thiabendazole and tridemorph. The fungus was found to be highly sensitive to carbendazim and thiabendazole even at 0.006 per cent.

Sandipan *et al.* (2007) reported that carbendazim significantly inhibited *M. phaseolina* at 100 ppm while non- systemic fungicide MEMC and mancozeb completely inhibited the growth of test fungus at all the concentrations.

Kumari *et al.* (2012) reported that *M. phaseolina* causes root rot of mungbean. Seven fungicides were tested *in vitro* by Poisoned food technique and in pots (*in vivo*). Bavistin was considered to be the most effective to inhibit mycelial growth of the pathogen as well as reducing root rot incidence, followed by captan, thiram, Indofil M-45 and Vitavax or Raxil, while copper sulphate was the least effective treatment in both the conditions.

Sangeetha and Jahagirdar (2013) reported that root rot/wilt causing pathogens are, *Sclerotium rolfsii*, *Rhizoctonia bataticola* and *Fusarium* sp. *In vitro* screening of fungicides was undertaken to identify an effective compound against these pathogens. Mancozeb, carbendazim, thiophanate methyl, hexaconazole, propiconazole, carbendazim + mancozeb and carboxin + mancozeb were found effective in inhibiting mycelial growth of all the three pathogens.

2.7.2 Evaluation of Bioagents

Mathur (1965) discussed associative effect of 13 soil saprophytes with 8 soil borne pathogens on Czapeks medium. *Trichoderma lignorum* (*T. viride*) and *Cunninghamella echinulata* inhibited all the pathogens by their rapid growth, the former also produced antagonistic metabolites. *M. phaseolina* and *Fusarium udum* were inhibited by 11 saprophytes.

Ghaffar (1968) reported that although 4 different types of reactions occurred between 46 soil isolates and *M. phaseolina* on culture plates, in most of the

cases the fungi were overgrown by *M. phaseolina*. *Trichoderma viride* however, grew and inhibited the growth of *M. phaseolina*, its hyphae coiling around those of latter.

Bashar (1991) reported that linear growth of *F. oxysporium* f.sp. *ciceri*, *F. solani*, *M. phaseolina* and *S. rolfsii* was completely inhibited at 5, 2.5, and 1 per cent concentrations, respectively of the *Clematis gouriana* extract filtered through a Seitz filter.

Cattelan (1994) evaluated 32 isolates of fluorescent *Pseudomonas* obtained from soybean rhizosphere and one unidentified isolate (B 183), obtained from soybean residues, *in vitro* for antagonism against *R. solani*, *S. rolfsii*, *Fusarium* spp., *M. phaseolina*, *Rosellina* spp. And *Sclerotinia sclerotiorum*. B 183 was the only isolate antagonistic to all the pathogenic fungi tested.

Sanchez *et al.* (1994) reported that the strain UPR5C of *Pseudomonas cepacia* strongly inhibited *M. phaseolina*, and also inhibited the growth of other *Phaseolus vulgaris* pathogens, including *S. rolfsii*, *R. solani*, *Pythium aphanidermatum*, and to a lesser extent *F. solani* in streak plate and double layer *in vitro* tests. they reported that inoculation of bean seeds with *Pseudomonas cepacia*, could be used as a seed treatment to protect against infection by root rot pathogens.

Perdomo *et al.* (1995) evaluated 64 strains of *Rhizobium* and seven other rhizosphere bacteria by streak plate, double layer and spent culture methods to determine their antagonistic activity against *M. phaseolina*, causal agent of ashy stem blight of beans. Results indicated that root nodule bacteria had weak biofungicidal potential. *Pseudomonas cepacia*, strain UPR5C consistently restricted fungal growth.

Selvarajan and Jeyrajan (1996) worked on inhibition of chickpea root rot pathogens, *F. solani* and *M. phaseolina* by antagonists and reported that *T. harzianum* induced maximum inhibition zone against *M. phaseolina*.

Ushamalini *et al.* (1997) reported that all the test antagonists (*T. harzianum*, *T. viride*, *T. hamatum*, *T. pseudokoningii*, *T. koningii*, *Bacillus subtilis* and *Pseudomonas fluorescens*) inhibited growth of *M. phaseolina*, significantly over control. Colony diameter of the pathogen was minimum (31mm) in *T. viride*, followed by *T. harzianum* (33mm) as against control (89mm).

Malathi and Doraisamy (2003) reported that *Trichoderma* spp., significantly inhibited the pathogen growth and the inhibition ranged from 32 to 67 per cent. The maximum inhibition was shown by *T. harzium*-5 followed by *T. koningii*.

Suriachandraselvan *et al.* (2004) tested *Trichoderma viride* and *Trichoderma harzanium* as potential antagonists to *Macrophomina phaseolina*. These two antagonists inhibited the mycelial growth and reduced the number, size and germination of sclerotia of *M. phaseolina*. Both *T. viride* and *T. harzanium* produced non-volatile metabolism in the culture filtrates, which showed the maximum fungitoxic effect against the pathogen. Further they conducted pot culture experiment to manage the charcoal rot of sunflower. Seed treatment with talc based formulation (4g/kg-1) of *Trichoderma viride* or *T. harzanium* significantly reduced the disease incidence (22-27 %) compared to untreated control (67 %).

Kumar *et al.* (2007) tested the efficacy of various isolates of *Pseudomonas fluorescens* for induction of systemic resistance against dry root rot of chickpea caused by *M. phaseolina* and reported that among of *M. phaseolina in vitro* and reduced root rot incidence under polyhouse condition.

2.7.3. Evaluation of organic amendments

Muthusamy and Mariappan (1992) reported that addition of organic amendment to soil activate the dormant propagules of pathogen to germinate and in absence of host, the germings may lyse before they produce new propagules. Oil cake application in soil reduced the propagules of *M. phaseolina* by stimulating the germination followed by lysis.

Hundekar *et al.* (1998) carried out *in vitro* evaluation of organic amendments against stalk rot of sorghum and reported that among the organic amendments tested neem, cotton, groundnut and safflower cakes were found effective in reducing inoculum levels of *M. phaseolina* and *Fusarium moniliforme*.

Sarvanan *et al.* (2004) reported that in green house experiments, all the organic amendments exhibited significant reduction in the population of the pathogen in the rhizosphere of banana. Among the amendments, neem cake showed the maximum reduction of population of the pathogen.

Suriachandraselvan *et al.* (2005) reported that among different organic amendments tested in pot culture experiment, FYM @ 12.5 tons/ha and neem cake @ 250 kg/ha were found effective in reducing the inoculum levels of *M. phaseolina* and significantly reduced the incidence of charcoal rot in sunflower to 13.3 and 15.0 per cent respectively from 63.3 per cent in control.

John *et al.* (2007) evaluated bioefficacy of native isolates of *Trichoderma virens* combined with FYM against dry root rot of urd bean incited by *M. phaseolina* under field condition. Seed treatment @ sowing @ 100g/kg + soil application (on 0 days after sowing and 30 days after sowing @ 100 g/m² of *T. virens* combined with FYM showed maximum reduction in dry root rot incidence.

George *et al.* (2007) reported that organic meal @ 1000-1500 kg/ha favoured better microbial population including rhizobia during crop period of groundnut.

2.8. Integrated management

Kaur and Mukhopadhyay (1992) reported that chickpea wilt complex can be effectively controlled by a biological agent *T. harzianum* and its integration with fungicides. Soil application with different doses of *T. harzianum* gave 53.50-85.70 per cent disease control in glass house condition. In the field integration of *T. harzianum* with different fungicides significantly reduced the incidence of chickpea wilt complex and increased crop yield. Seed treatment with Vitavax-200 and ziram resulted in 29.90 per cent disease control, which was further increased to 63.30 per cent when integrated with soil application of *T. harzianum*.

Mukhopadhyay *et al.* (1992) reported that integration of biological and fungicidal seed treatment i.e. treating seeds first with an antagonistic *Gliocladium virens* (10 conidia/ml) and then with fungicide 0.1 per cent has proven effective in managing several soil borne pathogens, including *S. rolfii*, *R. solani* and *Fusarium oxysporum* of chickpea, lentil and groundnut.

Vyas (1994) reported that simultaneous application of *T. viride* and *T. harzianum* with carbendazim to soybean seeds reduced dry root rot caused by *R. bataticola*. Yield of the treated plots was significantly higher (2.5 tonnes per hectare) as compared to check (1.4 t/ha).

Latha *et al.* (2000) conducted field trials on black gram to study the effect of antagonist and chemicals on root rot disease complex caused by *M. phaseolina* and cyst nematode (*Heterodera cajani*). Among nine treatments tested, combination of fungicides, carbendazim as seed treatment with carbofuran as soil application was most effective in reducing root rot incidence, nematode population and increasing pod yield. Among biocontrol agents *Paecilomyces lilacinus* with *T. viride* and *Pseudomonas fluorescens* as seed treatment resulted in reducing root rot incidence.

Mukherjee and Tripathi (2000) demonstrated biological and chemical control of wilt complex of french bean. It was reported that *Gliocladium virens* exhibited antagonistic effect against *S. rolfii*, *R. solani*, and *F. oxysporium* f.sp. *phaseoli*. Among 8 fungicides evaluated *in vitro*, Antracol was most effective against the pathogens. Seed treatment with spore mass of antagonist at 0.1 per cent and fungicides Opus, Contaf, and Antracol (0.1 per cent) increased seed germination, shoot length, vigour of plant and number of leaves per plant under glass house condition.

Deepak kumar and Dubey (2001) reported that seeds treated with captan (1g/kg)+*T. harzianum* (106 spores/ml/10g seed) gave good germination, least disease incidence along with highest green pod yield, which was statistically similar with seed treatment with *T. harzianum* alone, thiram + *T. harzianum*, carboxin + *Gliocladium virens* and *G. virens* alone.

Pant and Mukhopadhyay (2001) reported that integration of Vitavax (0.1 per cent) with both *G. virens* and *T. harzianum* improved seedling emergence, plant stand and yield significantly over individual components and combination of vitavax with *G. virens* proved the best treatment under field conditions in both the years. The highest increase in grain yield i.e. 114.14 per cent in 1995 and 144.11 per cent in 1996 was recorded when *G. virens* 0.2 per cent was integrated with Vitavax @ 0.1 per cent.

Prajapati *et al.* (2002) reported that carbendazim 25 DS, carbendazim 50 WP + thiram 80 WP, carboxin 75 WP and Topsin-M-50 WP completely inhibited the growth of *R. bataticola* and were found to be most effective.

Patil *et al.* (2003), conducted a field experiment to assess the efficacy of biocontrol agents viz., *Trichoderma harzianum*, *Trichoderma viride* both @ 5g/kg seed and *Pseudomonas fluorescens* both @ 10g/kg seed and fungicides viz. thirum, captan and carbendazim each @ 3g/kg seed as seed treatment against root rot (*Rhizoctonia bataticola*) disease of cotton for these successive crop seasons. Seed treatment with all the biocontrol agents and fungicides significantly reduced root rot incidence and also significantly increased yield of seed cotton(except carbendazim).

2.9 Screening

Agarwal and Sarbhoy (1976) evaluated soybean germplasm for resistance against *Macrophomina phaseolina*, and reported that sixty-four lines were found to moderately resistant and four lines viz., EC-1077, EC-25166, EC-39069 and EC-39797 as resistant under artificially inoculated field as well as glass house condition.

Singh *et al.* (1982) reported that 3 chickpea cultivars did not related to their resistance to *R. bataticola* Carbohydrate content was higher in the susceptible cultivar than in the 2 resistant ones.

Singh *et al.* (1989) studied sesamum germplasm for resistance against *Macrophomina* root rot and reported that out of 312 cultivar lines tested 30 lines were found resistant; 158 moderately resistant; and 124 showed susceptible reactions. The resistant lines were AK-77, ES-4, IS-175, KIS-430, N66-3169, SI-1740, SI-1153, SI-1875.

Reddy *et al.* (1990) reported additional sources of resistance to wilt and root rot in chickpea from ICRISAT, Hyderabad. Out of which 1883 accessions evaluated, 177 accessions showed less than 20 per cent mortality in the first year of screening and of these 40 accessions showed less than 10 per cent mortality in second year of screening.

Lodha and Solanki (1993) studied inheritance of resistance to dry root rot in cluster bean and reported that the susceptibility in the genotypes varied from 22.3 to 46.6 per cent. However, least (26.1per cent) pre-emergence mortality was recorded in the HFG-156 followed by IC-922913 (33.9 per cent) while Kutch-8 recorded lowest (22.3 per cent), post emergence mortality followed by RGC-471

Highly susceptible varieties HG-75 (40.5 per cent) and AG-111 (46.6 per cent) were identified for their use in resistant breeding programme.

Zote *et al.* (1993) screened chickpea germplasm lines against wilt and dry root rot pathogen and reported that BCP-28, BCP-29 and BCP-36 lines were resistant to *Fusarium* wilt and lines BCP-15, BCP-23, BCP-29, and BCP-30 were resistant to both wilt and dry root rot.

Gupta (1995) identified chickpea genotypes with dual resistance against wilt and root rot. He found that mortality in susceptible check JG -62 ranged from 80-100 per cent. Genotypes ICCV-89402, ICCV-90201, ICCV-90254, BG-209, H88-2, GF88426 and RSG-180 suffered less than 10 per cent incidence of both the diseases.

Sajeena *et al.* (2004) studied on screening of black gram varieties against root rot caused by *Macrophomina phaseolina* and reported that among the seventeen genotypes of black gram tested BG ADT 5 and BGKI recorded the maximum germination and yield and root rot disease incidence was at a minimum. IC14520 and IC14691 registered the minimum germination and yield and showed maximum disease incidence. None of the genotypes were found to be completely resistant to root rot disease.

Dinkaran *et al.* (2005) studied on screening of sesame genotypes against root rot pathogen *Macrophomina phaseolina* and reported that among the 1300 sesame genotypes screened only three accessions viz., RM-7, ORM-14 and ORM-17, showed resistant reaction.

Jayalakshmi *et al.* (2008) screened different genotypes for their resistance to dry root rot in pot culture experiment with susceptible check JG-62 during *rabi* season of 2004-2005 and 2005-2006. Out of 12 genotypes tested against root rot disease, 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot.

Aghakhani *et al.* (2009) stated that twenty – three isolates of *R. bataticola* causing dry root rot of chickpea (*Cicer arietinum*) collected from 10 different major chickpea growing states of India were highly variable in their morphological and cultural characters as well as pathogenicity /virulence. On the basis of the virulence analysis of the isolates on a set of chickpea cultivars namely

ICC12441, ICC1224, ICC12450, Pusa 362, BGD112, Pusa1103, Pusa212, Pusa1088 under blotter paper as well as sick soil grouped them into 6 pathotypes. The pathotype groups were related to agro ecological region of the country. The most virulent isolate (RBI from Bangalore, Karnataka) was fast growing and produced largest sclerotia. A set of cultivars was proposed for the first time for differentiating the pathotypes of *R. bataticola* causing dry root rot of chickpea.



Materials and Methods



CHAPTER –III

MATERIALS AND METHODS

3.1 Experimental materials

The following materials were used during the course of present investigation.

3.1.1 Disease samples

Plants showing typical symptoms of chickpea dry root rot were collected from farmers fields belonging to Marathawada region.

3.1.2 Culture medium

Common laboratory medium i.e. Potato Dextrose Agar (PDA) was used for isolation of the pathogen from infected plant samples.

Composition of medium

Potato Dextrose Agar Medium

Peeled potatoes	200.00g
Dextrose	20.00g
Agar	20.00g
Distilled water	1000 ml

3.1.3. Preparation of sand maize medium:

The sand maize medium with the 1:1 ratio of sand and maize, thoroughly mixed with 50 ml of water for every 20 g was used for mass multiplication of the pathogens.

3.1.4. Seeds

Genetically pure seeds of chickpea obtained from College of Agriculture, Badnapur, Agricultural research station Badnapur, and MPKV, Rahuri.

3.1.5. Soil

Medium deep black soil was collected from the experimental field of Department of Plant Pathology, College of Agriculture, Badnapur.

3.1.6. Chemicals

The chemicals, ingredients, reagents etc. used for the preparation of media were obtained from the Department of Plant Pathology, College of Agriculture, Badnapur.

3.1.7. Sand

Clean sand was collected from the college of Agriculture, Badnapur.

3.1.8. Biological agents

The isolates of *T. viride*, *T. harzianum*, *T. hamatum*, *T. longibrachiatum*, *T. koningii*, *T. virens* and *Aspergillus niger*, were obtained from the Department of Plant Pathology, College of Agriculture, Parbhani..

3.1.9. Fungicides

The following fungicides were used *in vitro* studies under laboratory condition against *Rhizoctonia bataticola*.

Sr. No.	Common name	Chemical name	Trade name	Source
1	Propiconazole	1-2-(2,4-D)-4- Propyl-1,3dioxolan-2yl methyl IH - 1,2,4, Triazole	Tilt 25% EC	Synjenta Agrochemicals
2	Carbendazim	Methyl 2-Benzimidazole carbamate	Bavistin 50 WP	BASF India Ltd. Mumbai
3	Captan	N-(Trichloro Methylthio) 4-cyclohexane-1,2 dicarboximide	Captan 75% WP	Rallis India, Pvt. Ltd. Mumbai
4	Carboxin	5,6-dihydro 2-methyl, 1,4-oxathin, 3 carbonate	Vitavax 75 EC	Hindustan Insecticides Ltd.
5	Benomyl	Methyl-N (1 Butyl carbonyl) 2-Benzimidazole carbonate	Benlate 50WP	Corromandel Fertilizer Ltd.
6	Mancozeb	Manganese ethylene bis dithiocarbonate plus zinc	Indofil M-75 EC	Indofil Chemicals Ltd. Mumbai
7	Thiram	Tetramethyl thiuram disulphide	Thiram 75%	Bharat Pulverising mills Ltd., Mumbai.
8	Carbendazim + mancozeb	Methyl 1H-benzimidazole-2-yl carbamate + Manganese ethylene bis-dithiocarbamate plus zinc	Saff 75% WP	
9	Control	-	-	

3.1.10. Field and laboratory facilities

All field experiments were planned and conducted on the research farm of the Department of Plant Pathology, College of Agriculture, Badnapur and Agricultural Research Station, Badnapur. *In vitro* studies were conducted in the laboratory and screen house of plant pathology department.

3.1.11. Glasswares

Different types of Corning and Borosil brand glasswares were used in the present investigations. The common glasswares used were Petriplates, test tubes, conical flasks, measuring cylinder, glass rod, spirit jar, beakers and pipets.

3.1.12 Equipments

The laboratory equipments were used such as Autoclave, B.O.D. incubator, Refrigerator, Shaker, Research microscope, and analytical scientific balance etc.

3.1.13 Miscellaneous materials

Inoculating needle, forceps, blotter paper, spirit lamp, mercuric chloride, cork borer, plastic bucket, distilled water, earthen pots, paper bags, polythene bags, labels, scales, soil isolation chamber, laminar airflow cabinet, were used during the course of present investigations were available in the Plant pathology section, College of Agriculture, Badnapur, and Agricultural research station, Badnapur.

3.2 Methodology

3.2.1 Survey

A field survey of chickpea dry root rot disease was conducted in eight districts of Marathawada region during the months of December and February, 2015-16 to record occurrence and distribution of chickpea dry root rot disease. On an average 10 farmers fields of chickpea in each district were visited and the percent dry root rot disease incidence was recorded by counting total chickpea plants in 1x1m² area and total dry root rot infected plants in that area. Chickpea plants

showing typical symptoms were collected in separate paper bags and were brought to the laboratory for investigations. The intensity of disease was noted by counting at least 400 plants in each field. Observation on dry root rot intensity was recorded as, the symptoms expressed as straw colored plant like dry grass at pod formation , black rotted roots , shredding of bark and roots broken easily, minute dark black sclerotial bodies on root surface and in the pith region.(PLATE -)

The per cent disease incidence or intensity was calculated by using 0-9 disease rating scale(AICRP scale). On an average ten villages of each district were surveyed for dry root rot severity are presented in below table.

List of Chickpea growers in Marathwada region

Sr. No	Name of farmer	District	Village	Stage of crop	Variety
1	Prabhakar Pawar	Aurangabad	Pandhari	Podding	Vijay
2	Ganesh Pawar		Goaltagaon	Podding	Vijay
3	Appasaheb Jadhav		Itkheda	Flowering	LocalS
4	Sainath Jadhav		Karmad	Podding	BDN 797
5	Devidas Kite		Bokudjalgaon	Flowering	Digvijay
6	Pandurang Pawar		Lasur	Podding	Virat
7	Uttam Madake		Chaukha	Podding	Vijay
8	Sanjay Tupe		Shendra	Podding	Virat
12	Maruti Totare	Nanded	Naigaon	Flowering	local
13	Khandu Totare		Vishnupuri	Flowering	Local
14	Rama Totare		Takalgao	Flowering	Divijay
15	Eknath Totare		Ardhapur	Podding	Vijay
16	Dattatraya Gaikwad		Dabad	Podding	Vijay
17	Ashokrao Rodage	Parbhani	Parbhani	Podding	BDN 797
18	Pratap Rodage		Pokharni	Podding	BDN 797
19	Shivaji Rodage		Assola	Podding	Virat
20	Suresh Rodage		Purna	Podding	Local
21	Bhaskar Rodage		Manwat	Flowering	vijay
22	Sachin Kadam		Gunj	Flowering	Digvijay
23	Shriram Khandare		Dharasur	Flowering	Local
24	Ganeshrao Deshmukh		Daithna	Flowering	Digvijay
25	Ramesh Bhopale	Jalana	Kandhar	Podding	Digvijay
26	Rameshvar Kankhar		Badnapur	Podding	Virat
27	Baban Jadhav		Assola	Podding	Vijay
28	Ganesh Bhopale		Kondi	Podding	BDN-797
29	Ambadas Kanhere		Ranjni	Podding	Local
30	Ramesh Lahane		Sawewadi	Podding	Local
31	Prabhakar Joshi		Viduli	Podding	Digvijay
32	Appasaheb Kanhere		Jaypur	Podding	Virat

33	Sanjivani Pandule	Beed	Bhavthana	Flowering	Local
34	Sagar Pandule		Bardapur	Podding	Virat
35	Asurba Mutkule		Anandgao	Podding	Digvijay
36	Vijay Pokale		Ladegao	Podding	Vijay
37	Ajinath Pandule		Kumbephal	Podding	Vijay
38	Ashabai Pandule		Jaygao	Podding	BDN-797
39	Rahul Jagatap		Jirewadi	Podding	Vijay
40	Shivajirao Mane	Latur	Sai	Flowering	Vijay
41	Vasantarao Patil		Arvi	Flowering	Vijay
42	Ambadas Nade		Kasarkheda	Flowering	Virat
43	Vivek Shelke		Janwal	Podding	Virat
45	Babasaheb Hudage		Sugaon	Podding	Virat
47	Ravasaheb Patil		Garsoli	Podding	Digvijay
49	Shakarrao Sabade		Halli	Podding	Virat
50	Sadashiv Futane	Osmanabad	Ter	Flowering	vijay
51	Mamasaheb Futane		Upla	Flowering	Vijay
52	Vishal Bhore		Chinchola	Podding	Vijay
53	Sudhir Sagare		Songiri	Podding	Vijay
54	Mahadeo Rote		Ramwadgao	Podding	Virat
55	Bhanudas Tate	Hingoli	Salesuea	Podding	Virat
56	Vinod Tate		Hiwara	Podding	Virat
57	Gajanan Tate		Nandkheda	Podding	Local
58	Rohidas Mhase		Purna	Podding	Local
59	Ramesh Deokate		Hatta	Podding	Vijay

3.2.2 Isolation, pathogenicity, reisolation, symptomatology

3.2.2.1 Isolation.

Chickpea plants, showing typical dry root rot symptoms were collected from Agricultural Research station Badnapur. The roots of infected plants were washed in tap water to remove soil adhered to roots. The roots were cut into small bits (2.5mm) with sterilized blade. These bits were then disinfected with 0.1 per cent aqueous solution of mercuric chloride (HgCl_2) for two minutes and washed with three subsequent changes of sterilized water to remove traces of mercuric chloride. These bits were sterilized blot dried and four bits per petriplate were placed on the sterilized solidified potato dextrose agar (PDA) plates. These plates were then incubated at $28 \pm 2^\circ\text{C}$ temperature for seven days. The well isolated fungal growth, free from any contamination was transferred to agar slant by hyphal tip method. By frequent sub culturing, it was purified and agar slants showing pure fungal growth were maintained and used for further studies.

3.2.2.2. Pathogenicity

The full grown inoculum of *R. bataticola* multiplied on sterilized sand maize medium was mixed @50 g/kg soil in each sterilized earthen pot of 18 cm diameter containing sand and soil (1:1) Pot simply sown without inoculums was maintained for reference purpose as control. The inoculated pot was moistened with water and incubated for 15 days. The surface sterilized seeds of susceptible chickpea variety JG- 62 were sown @10 seeds per pot. The pots were kept in glass house and observations were recorded on root rot incidence for confirmation of pathogenicity.

3.2.2.3. Reisolation

The reisolations were made on PDA plates from roots of artificially diseased plants showing typical symptoms of dry root rot. The fungal growth obtained was transferred on potato dextrose agar slants for comparing with original culture.

3.2.2.4. Symptomatology

The symptoms of dry root rot were observed at flowering and pod filling stages till complete drying of plants in artificially inoculated soil and under field conditions.

3.3. Cultural studies

The cultural characters of the pathogen were studied on the following solid media viz.,

3.3.1. Semi-synthetic media

1. Potato Dextrose Agar

Semi synthetic media the composition and preparation of different media are given below.

Peeled Potatoes	-	200 g
Dextrose	-	20 g
Agar-agar	-	20 g
Distilled water	-	1000 ml

Potato slices were boiled in 500 ml distilled water for 20 minutes and filtered through muslin cloth. Agar-agar was melted in another 500 ml distilled water and

dextrose 20 g was added. Both were mixed and the volume was made up to 1000 ml and autoclaved at 121°C for 15 lbs pressure for 15 minutes.

2. Host extract agar (chickpea)

Healthy chickpea leaves	-	200 g
Dextrose	-	20 g
Agar-agar	-	20 g
Distilled water	-	1000 ml

Chickpea leaves were boiled in 500 ml of water for 30 minutes. Extract was collected by filtering through muslin cloth. The agar-agar was melted in 500 ml of water and dextrose 20 g was added. Both the solutions were mixed and the volume was made to 1000 ml and sterilized.

3. Oat meal agar

Oat meal	-	30 g
Agar-agar	-	20 g
Distilled water	-	1000 ml

Oat meal was boiled in 500 ml distilled water for 30 minutes, filtered through muslin cloth; while agar-agar was melted in 500 ml distilled water. Two solutions were mixed and the volume was made to 1000 ml and then sterilized.

4. Carrot agar

Carrot	-	200 g
Dextrose	-	20 g
Agar-agar	-	20 g
Distilled water	-	1000 ml

Carrots were peeled off and cut into small pieces and then 200 g of peeled pieces were boiled in 800 ml of water and extract was filtered through muslin cloth. The dextrose was dissolved into the solution and agar-agar was added and boiled until it was properly dissolved Further volume was made up to 1000 ml by adding distilled water.

5. Pline agar

Dextrose	-	20 g
Agar-agar	-	20 g
Distilled water	-	1000 ml

Agar-agar was melted in 500 ml distilled water and dextrose 20 g was added. Both were mixed and the volume was made up to 1000 ml and autoclaved at 121°C at 15 lbs for 15 minutes.

3.3.2 Synthetic media

1. Richard's agar

Potassium nitrate (KNO_3)	-	10 g
Potassium monobasic phosphate (KH_2PO_4)	-	5 g
Magnesium sulphate ($\text{Mg SO}_4 \cdot 7\text{H}_2\text{O}$)	-	2.5 g
Ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$)	-	0.02 g
Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)	-	50 g
Agar-agar	-	15 g
Distilled water	-	1000 ml

Agar was melted in 500 ml distilled water and rest of the ingredients were thoroughly dissolved in 500 ml distilled water. Both the preparations were mixed and the final volume was made up to 1000 ml and then autoclaved.

2. Czapek's Dox agar

Sodium nitrate (NaNO_3)	-	3 g
Potassium dihydrogen phosphate (K_2HPO_4)	-	1 g
Magnesium sulphate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$)	-	0.5 g
Ferrous sulphate ($\text{FeSO}_4 \cdot 7 \text{H}_2\text{O}$)	-	0.19 g
Sucrose ($\text{C}_{12} \text{H}_{22}\text{O}_{11}$)	-	30 g
Agar-agar	-	15 g
Distilled water	-	1000 ml

Agar was melted in 500 ml distilled water and rest of the ingredients were thoroughly dissolved in 500 ml distilled water. Both the preparations were mixed and the final volume was made up to 1000 ml and then autoclaved.

3. Elliot's Agar (EA)

Sodium carbonate (Na_2CO_3)	-	1.05 g
Magnesium sulphate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$)	-	0.60 g
Asparagine	-	3.00 g
Dextrose ($\text{C}_6\text{H}_{12}\text{O}_6$)	-	3.00 g
Potassium dihydrogen orthophosphate (KH_2PO_4)	-	1.36 g
Agar-agar	-	20.00 g
Distilled water	-	1000 ml

Agar-agar was melted in 400 ml distilled water. The other ingredients were dissolved in 400 ml of distilled water. The two solutions were mixed thoroughly and the volume was made up to 1000 ml by adding distilled water.

4. Sabouround's agar

Dextrose	-	200 g
Peptone	-	20 g
Agar-agar	-	20 g
Distilled water	-	1000 ml

Dextrose, peptone and agar-agar were dissolved in the distilled water by boiling them together.

3.3.3. Solid media test

The agar agar based sterilized media were poured aseptically into 9.0 cm dia. previously sterilized Petri plates @ 20 ml plate⁻¹. The Petri plates were inoculated aseptically after solidification by placing 5 mm dia. mycelial disc of 10 days old pure culture of *Rhizoctonia bataticola* maintained on PDA medium at the centre. Three repetitions were kept for recording growth and one set was kept for recording sporulation in each case. The Petri plates were incubated at $27 \pm 1^\circ\text{C}$ temperature in BOD incubator. The radial growth was daily measured till the plates were covered with the fungus mycelium and colour of the fungus on different media was recorded by using standard colour chart. The data thus, obtained were statistically analyzed.

3.3.4. Measurement of sporulation

The spore suspension was prepared and observed under microscopic field for counting the number of spores per microscopic field (five times) and average number of spores/ microscopic field (MF) was worked for each treatment. The degree of sporulation was graded as below.

Table 2: Grading of sporulation

No.	Grade	Spores/MF
1	Nil	No spores
2	Poor	1-50
3	Moderate	51-100
4	Good	101-150
5	Excellent	> 151

MF= Microscopic field

3.3.5. Morphological studies

Slide culture technique was employed to obtain details of morphological characters.

3.4 Disease management

3.4.1. *In-vitro* studies

3.4.1.1. Evaluation of fungicides against *R. bataticola*

The effect of nine fungicides on *R. bataticola* (Taub) Butler was evaluated *in-vitro* by employing "Poisoned food technique". The requisite amount of each fungicide @ 500, 1000 and 2000 ppm based on active ingredient was added in an autoclaved and cooled (45°C) potato dextrose agar medium to obtain desired concentrations. The fungicides used with their concentrations were propiconazole, carbendazim, captan, carboxin, benomyl, mancozeb, thiram, carbendazim + mancozeb, Plain PDA without fungicide served as control. The PDA was then poured into sterile glass petriplates (90 mm diameter) and allowed to solidify. These plates were inoculated with a 5 mm mycelial disc of the test fungus and incubated for 7 days at 28 ± 2°C. All the treatments were replicated thrice. Observations on radial mycelial

growth in treatment plates were recorded after 7-8 days .Per cent inhibition of the test fungus over control was calculated by applying the formula given below.

Details of the experiment :-

Design – Complete Randomized Design (CRD)

Replications – Three

Treatments – Nine

T1 – Propiconazole

T2 - Carbendazim

T3 - Captan

T4 - Carboxin

T5 - Benomyl

T6 - Mancozeb

T7 - Thiram

T8 - Carbendazim + mancozeb

T9 – Control

$$\text{Per cent inhibition (I)} = \frac{C - T}{C} \times 100$$

Where,

C = Growth (mm) of test fungus in control plate.

T = Growth (mm) of test fungus in treatment plate.

3.4.1.2. Evaluation of biocontrol agents against *R. bataticola*

Seven biocontrol agents were assessed *in vitro* against *R. bataticola* by dual culture method. Mycelial discs of 5 mm diameter were cut from the margin of 7 days old culture of both the test pathogens and placed opposite to each other on sterilized PDA in petriplate. The inoculums discs were placed 30 mm away from each other. The petriplate inoculated with disc of *Rhizoctonia bataticola* alone served as control. The inoculated plates were incubated at $27 \pm 2^{\circ}$ C in BOD incubator for 7 days. The per cent growth inhibition of inserting colonies was calculated as per formula given below.

Details of experiment :

Design – CRD

Replications – Three

Treatments – Eight

T1 – *Trichoderma viride*

T2 – *Trichoderma harzianum*

T3 - *Trichoderma hamatum*

T4 - *Trichoderma longibrachiatum*

T5 - *Trichoderma. koningii*

T6 – *Trichoderma virens*

T7 – *Aspergillus niger*

T8 - Control

$$\text{Per cent inhibition (I)} = \frac{A_1 - A_2}{A_1} \times 100$$

Where,

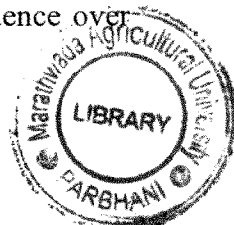
A_1 = Area covered by pathogen in control plate.

A_2 = Area covered by pathogen in dual culture plate

3.5 Pot culture experiment

3.5.1 Evaluation of organic amendments against *R. bataticola*

A pot culture was conducted during *Rabi* season of 2015-16 at green house, Department of Plant Pathology, College of Agriculture, Badnapur. An experiment was laid out in CRD with three replications and 7 treatments. The soil in pot was made sick by sand maize medium. To this sick soil, each organic amendment was added @ 50g and 100 g in each pot. These each pot were again incubated for 8 days. After incubation, pots were sown with 5 seeds per pot and kept in glass house. Each treatment was replicated three times. Per cent disease incidence (PDI) was recorded at 30, 45, 60, 75, 90 days after sowing. The per cent disease incidence over control was calculated.



Details of experiments :

Design – CRD

Replications – Three

Treatments – Seven

T1 – FYM

T2 – Castor cake

T3 – Vermicompost

T4 – Poultry manure

T5 – Mustard cake

T6 – Neem seed cake

T7 - Control

$$PDI = \frac{\text{DI in untreated seeds} - \text{DI in treated seeds}}{\text{DI in untreated seeds}} \times 100$$

3.5.2. Integrated management of dry root rot of chickpea in pot culture

A pot culture experiment was conducted in the screen house at Department of Plant Pathology during *Rabi* season of 2015-16 at College of Agriculture, Badnapur, to integrate those fungicides, bioagents and organic amendments found most effective against *R. bataticola* during present *in vitro* studies.

The pot culture experiment was laid out in CRD with three replications and 9 treatment using susceptible JG-62 cultivar to *R. bataticola*. Plants showing symptoms were considered as diseased and per cent disease incidence was recorded at 30,45,60,75 and 90 days after sowing. The per cent disease incidence over control was calculated.

Details of experiments:

Design – CRD

Replications – Three

Treatments – Nine

- T1 – Carbendazim
- T2 – Benomyl
- T3 – Carbendazim + mancozeb
- T4 – *Trichoderma viride*
- T5 – *Trichoderma harzianum*
- T6 – *Trichoderma hamatum*
- T7 – Neem seed cake
- T8 – FYM
- T9 - Control

$$\text{(PDI)} = \frac{\text{DI in untreated seeds} - \text{DI in treated seeds}}{\text{DI in untreated seeds}} \times 100$$

3.6. Screening technique

Available germplasm lines, genotypes cultivars/varieties of chickpea were evaluated for their reaction against *R. bataticola*. The experiment was conducted during Rabi 2015-16 under green house condition in pots.

Twenty five chickpea genotypes/lines obtained from ARS, Badnapur, VNMKV, Parbhani were screened against *R. bataticola* in pots by using sick soil technique under green house conditions.

To carry out the screening, earthen pots were taken with sterilized soil and kept in screen house. Five seeds of each cultivar were sown in each pot and later two seedlings per plot were maintained for screening purpose. Three pots per entry per replication were maintained. Soil was made sick by applying culture of *R. bataticola*. After inoculation, the pots were watered regularly. Observation were recorded on after inoculation

Based on the disease incidence, test lines were categorized for their reaction to dry root rot as per rating scale given by Nene (1981).

Disease rating scale;

Table: Disease reaction category based on disease incidence

Disease incidence (%)	Disease reaction
1.0-10.0 %	Resistant
10.1-20.0%	Moderately resistant
20.1-30.0%	Moderately susceptible
30.1-50.0%	Susceptible
Above 50 %	Highly susceptible

3.7. Statistical analysis

The experimental data was statistically analyzed at computer centre, College of Agriculture, Badnapur. Observation of per cent disease incidence and intensity were converted to values of arcsine transformation for statistical analysis.



Results and Discussion



CHAPTER IV

RESULTS AND DISCUSSION

Chickpea (*Cicer arietinum* L) is an important pulse crop of India popularly known as 'Gram'. In India, it is cultivated on nearly about 10.22 million ha. area with an annual production of 09.88 metric tonnes and productivity of 967 kg/ha. Maharashtra contributes 18.19 lakh ha. area with production of 16.22 lakh tonnes and productivity of 891 kg/ha in *Rabi*. The Marathwada region contributes to 6.58 lakh ha. area with production of 06.64 lakh tonnes and productivity of 918 kg/ha, during *Rabi* (Anonymous, 2014).

Among the various factors responsible for low production and productivity of chickpea, diseases caused by biotic agents viz., fungi, bacteria, viruses and nematodes are the major ones. Of the major fungal diseases infecting chickpea, dry root rot incited by *Rhizoctonia bataticola* (Taub) Butl. is one of the most destructive and wide spread disease which cause average yield losses of 05-50% (Anonymous, 2012).

Keeping in view, economic importance of chickpea and losses incurred by dry root rot disease, present investigations on the various aspects viz., survey, isolation, identification and pathogenicity, effect of culture media, *in vitro* evaluation of fungicides, bioagents , organic amendments, integrated disease management, and germplasm screening against *R. bataticola* were undertaken during *Rabi*, 2015-16 at the Department of Plant Pathology, College of Agriculture, Badnapur and Agriculture Research Station, Badnapur(VNMKV Parbhani). The results obtained on the above aspects during the present investigations are interpreted in the following paragraphs.

4.1. Survey

A survey for the incidence of dry root rot of chickpea *R. bataticola* was carried out in farmers chickpea crop fields comprising different eight districts of Marathwada region of the Maharashtra state during *Rabi*, 2015-16. The chickpea fields under rainfed condition were surveyed at maturity stage and the results obtained are presented in Table 1 and Fig 2.

4.1.1. Status of dry root rot in Marathwada :-

Data presented in Table 1 showed that incidence of dry root rot at maturity stage in Marathwada region varied from 06.02-25.02 per cent. The highly prone areas of dry root rot of chickpea were Amla (25.02%), followed by HalliHANDARGULI (24.30%), BORGAEON (23.90%), GARSOLI (23.60%), JAYPUR (23.10%) and SUGAEON (23.00%). The least incidence of disease was noticed in DABAD (06.02%), as compared to other locations surveyed.

4.1.2. District-wise seasonal incidence

Results (Table 1, PLATE I and Fig.1) revealed that during *Rabi*, 2015-16, average of dry root rot incidence in the eight districts surveyed was ranged from 10.20 (Aurangabad) to 23.00 per cent (Latur).

Table 1. District-wise incidence of chickpea dry root rot disease during *Rabi*, 2015-16

Districts	Tahsils	Villages	Chickpea varieties/cultivars	Incidence (%)
Jalna	Badnapur	Kandhar	Digvijay	20.10
		Badnapur	Virat	20.90
		Assola	Vijay	21.25
	Partur	Kondi	BDN-797	21.40
		Ranjni	Local	21.70
		Sawewadi	Local	22.23
	Mantha	Viduli	Digvijay	22.80
		Jaypur	Virat	23.10
Average				21.68
Beed	Ambajogai	Bhavthana	Local	14.20
		Bardapur	Virat	14.90
	Kaij	Anandgaon	Digvijay	15.00
		Ladegaon	Vijay	15.35
		Kumbephal	Vijay	16.23
	Parli	Jaygao	BDN-797	16.80
		Jirewadi	Virat	17.00
	Average			
Latur	Latur	Sai	Vijay	20.00
		Arvi	Vijay	22.50
		Kasarkheda	Vijay	21.40
	Chakur	Janawal	Virat	22.90
		Sugaon	Virat	23.00
	Renapur	Garsoli	Virat	23.60
		Borgaon	Digvijay	23.90
	Ahamadpur	Hallihandarguli	Virat	24.30
		Amla	Digvijay	25.02
Average				23.00

Hingoli	Hingoli	Salesura	Virat	11.70
		Hiwara	Virat	12.00
		Nandkheda	Vijay	12.80
	Vasmat	Purna	Vijay	13.20
		Hatta	Vijay	13.82
		Adgaon	Vijay	14.02
	Aundha	Aundha	Local	14.45
Average				13.14
Osmanabad	Osmanabad	Ter	Vijay	10.01
		Upla	Vijay	11.17
	Bhoom	Chinchola	Vijay	12.18
		Songiri	Vijay	13.13
	Kalamb	Ramwadgao	Virat	13.60
		Charatha	Virat	13.99
		Tandalwadi	Virat	14.20
Average				12.62
Nanded	Nanded	Naigao	Local	14.00
		Vishnupuri	Local	14.40
		Takalgao	Digvijay	14.90
	Ardhapur	Ardhapur	Vijay	15.20
		Dabad	Virat	06.02
Average				14.90
Parbhani	Parbhani	Parbhani	BDN-797	16.60
		Pokharni	BDN-797	17.00
		Assola	Virat	17.30
	Purna	Purna	Local	17.80
	Manvat	Ambejao	Vijay	18.00
		Gunj	Digvijay	18.02
	Gangakhed	Dharasur	Local	18.64
		Daithana	Digvijay	19.00
Average				17.79
Aurangabad	Aurangabad	Pandhari	Vijay	09.00
		Goaltgaon	Vijay	09.08
		Itkheda	BDN-797	09.09
		Karmad	Digvijay	10.00
	Paithan	Bokudjalgaon	Virat	10.04
	Vaijapur	Lasur	Vijay	10.20
	Sillod	Chaukha	Virat	11.50
	Chikhalthana	Shendra	Local	11.20
Average				10.20
Overall Average	Total locations (59)			16.12

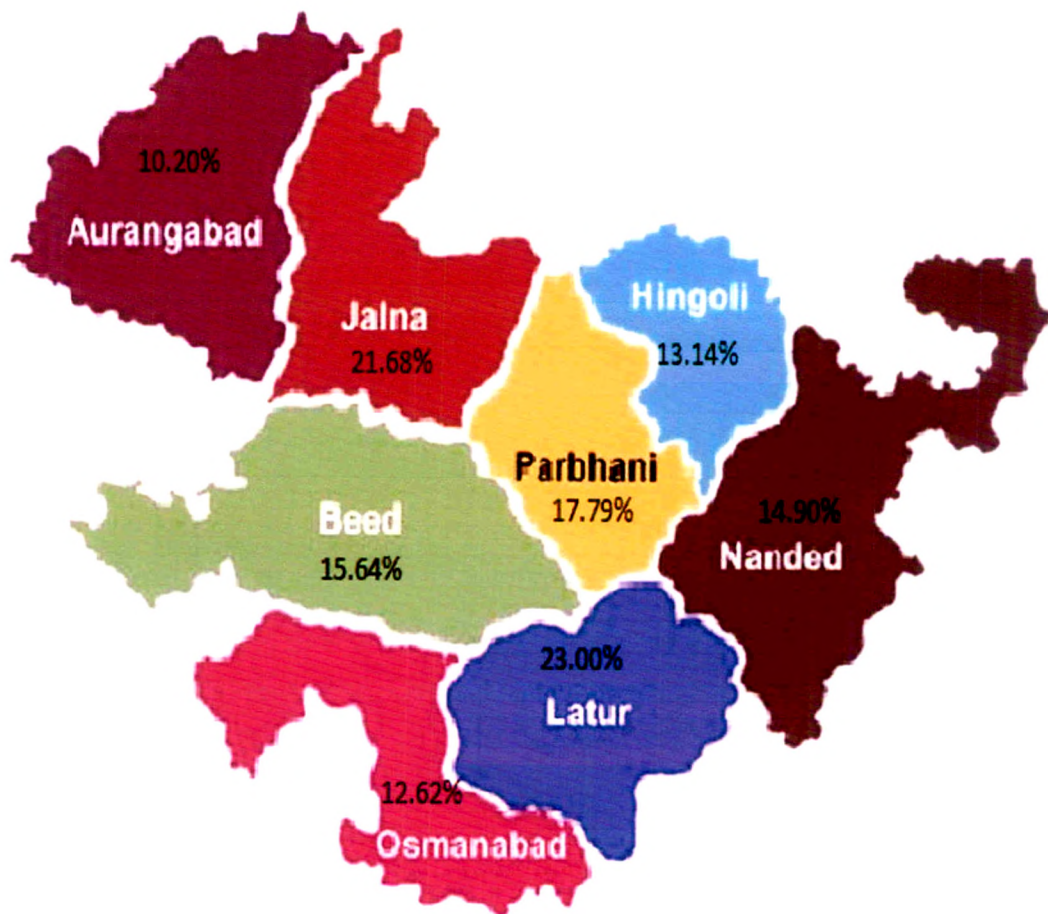


Fig. 1. Map of Marathwada region showing incidence of dry root rot of chickpea.

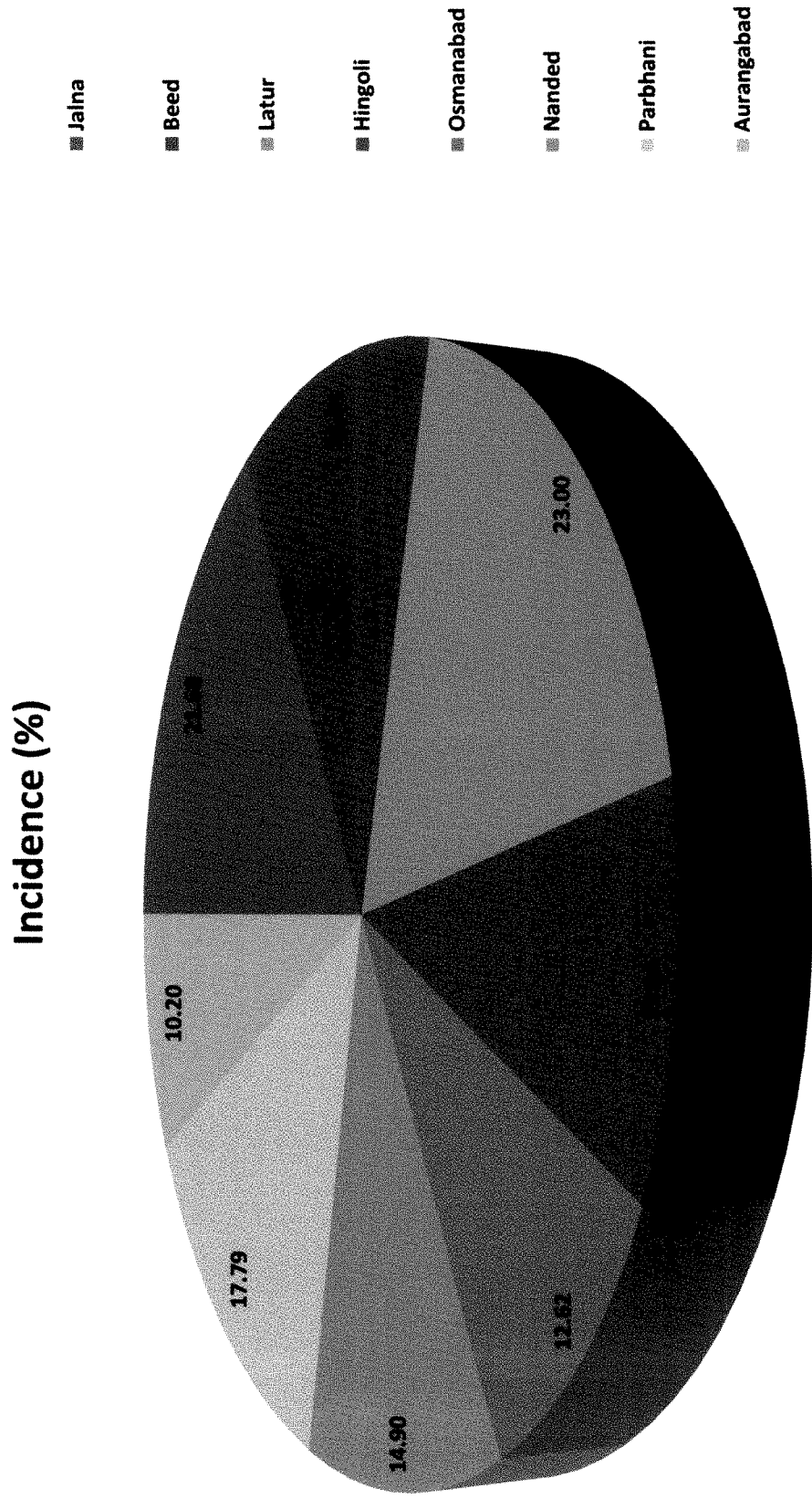


Fig. 2. District-wise incidence of chickpea dry root rot disease during Rabi, 2015-16.

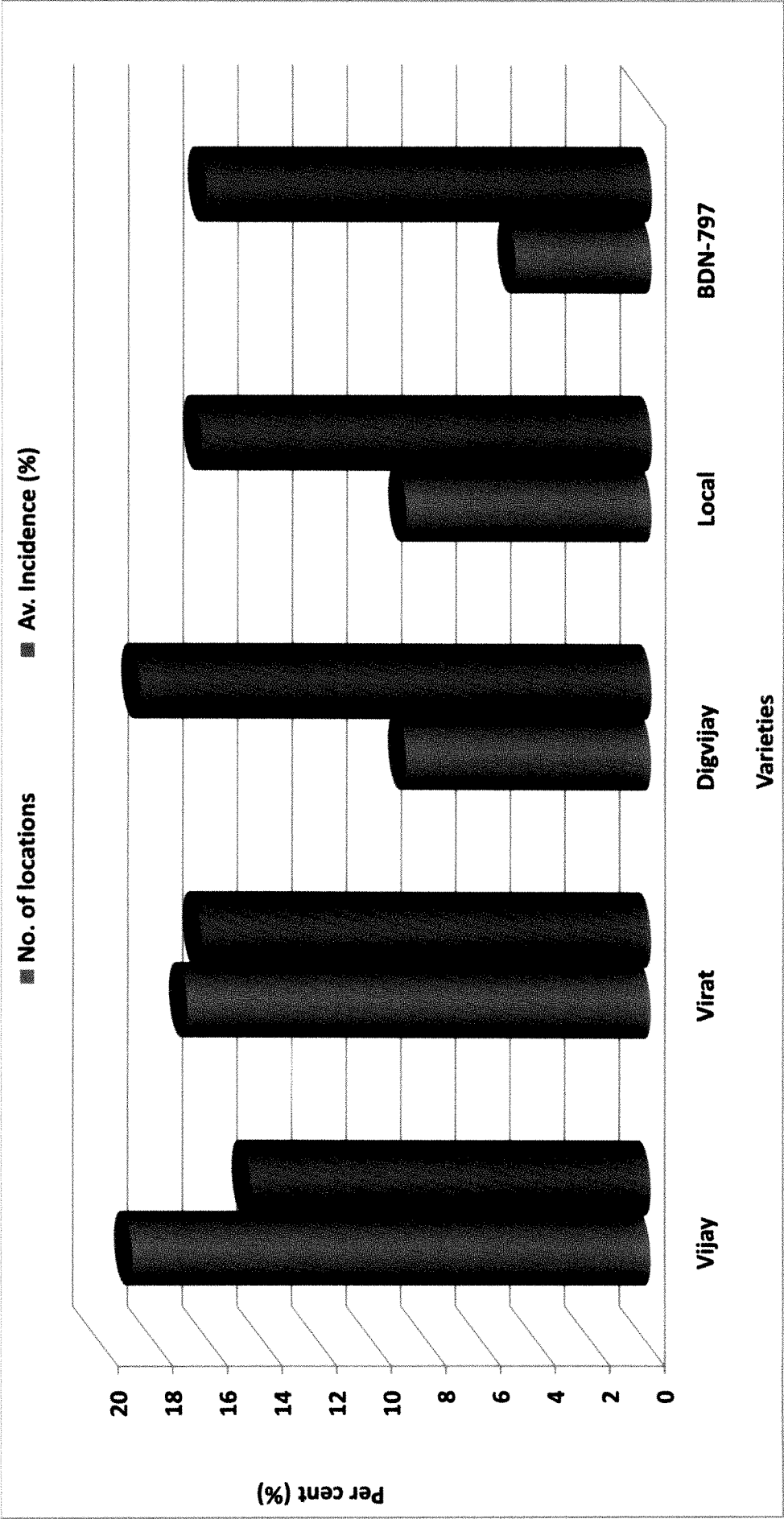
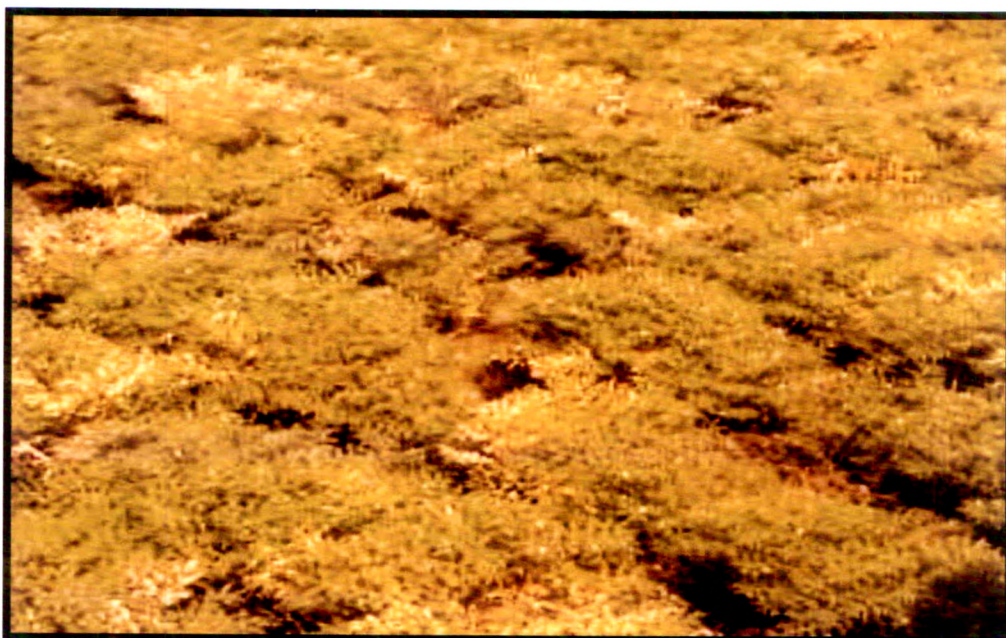


Fig. 3. Variety-wise incidence of chickpea dry root rot disease in the major districts of Marathwada during Rabi, 2015-16.

PLATE I



**view of Dry root rot of chickpea during survey
on farmers field**

However, the chickpea crop grown in the district of Latur was found to be infected maximum with an average of 23.00 per cent. This was followed by the districts viz., Jalna (21.68 %), Parbhani (17.79%), Beed (15.64%), Nanded (14.90%), Hingoli (13.14%), Osmanabad (12.62 %) and Aurangabad (10.20 %).

4.1.3. Variety-wise seasonal incidence

In eight districts of the Marathwada region surveyed, about 5 different chickpea varieties were grown by the farmers. The results obtained on variety wise incidence of dry root rot are presented in the Table 2 and depicted in the figure 2.

Table 2. Variety-wise incidence of chickpea dry root rot disease in the major districts of Marathwada during *Rabi*, 2015-16

Varieties/cultivars	No. of locations	Av. Incidence (%)
Vijay	19	14.70
Virat	17	16.50
Digvijay	09	18.76
Local	09	16.51
BDN-797	05	16.34

During *Rabi*, 2015-16 (Table 2 and Fig. 2) in Marathwada region the average dry root rot disease incidence on the chickpea varieties grown was ranged from 14.70 (Vijay) to 18.76 (Digvijay) per cent. However, maximum average dry root rot incidence (18.76%) was recorded on Digvijay cultivar. This was followed by the varieties/cultivars viz., Local (16.51%), Virat (16.50%), Vijay (14.70%) and BDN-797 (16.34%).

The results revealed that the per cent disease incidence varied from location to location and variety to variety. However, the dry root rot disease was found in almost all the chickpea fields surveyed. The overall average disease incidence was more in Latur district as compared to other districts surveyed. Where the Crop was grown in light to medium type of soil and moisture stress occurred at pod filling stage. The results of present finding are, Wakocha (2000) who stated that, the disease incidence of dry root rot soybean was significantly low when the

soil moisture was 60-70% but it was 0.5% at 10-20% soil moisture. Date (2015) conducted survey for recording dry root rot incidence of 15.75% ranged from 10.12 to 23.14%. District wise incidence of disease was Maximum in Latur (23.14%) followed by Jalna (22.2%).

Such variation in incidence and wide spread nature of chickpea have been reported by earlier workers like Maheshwari *et al.* (1983), Sharma *et al.* (1983), Pandey and Singh (1990), Srivastava *et al.* (2002), Manjunatha *et al.* (2011).

4.2 Isolation

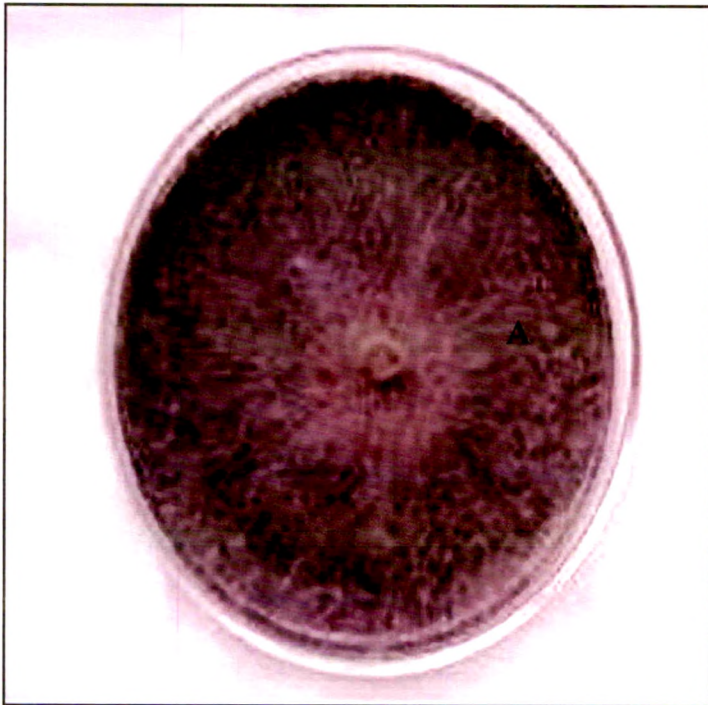
The test pathogen (*Rhizoctonia bataticola*) was isolated successfully on the basal culture medium potato dextrose agar (PDA), from the roots of infected chickpea plants showing typical symptoms of dry root rot, by standard tissue isolation method (PLATE –II) four days after inoculation. The pathogen produced dark brown to black colored colonies with septate mycelial growth on Potato dextrose agar (PLATE -II). Pure culture of *Rhizoctonia bataticola* was obtained by repeated sub culturing on PDA plates and maintained on PDA slant for further studies.

The present observations of the pathogens resemble to earlier reports on isolation of *R. bataticola* from infected chickpea plant carried out by Mitra (1931), Sharma and Khare (1969), Bhatti *et al.* (1987), Beniwal *et al.* (1992), Demirci *et al.* (1998), Sharma *et al.* (2004).

4.3 Pathogenicity test

The pathogenicity test was carried out by soil inoculation method which revealed that typical dry root rot symptoms developed after 30-40 days after inoculation, which supported to pathogenicity of isolate to JG-62 genotype. The plant started drying from base to top from flowering stage which confirmed that disease was advancing and finally whole plant completely dried. The disease was confirmed as plants were easily pulled out and further the necrosis symptoms were noticed on roots. The black near ground level observed shredded host plant. The blackish sclerotial bodies were also noticed on the exposed root particularly at collar region. The developed symptoms which were recorded resembled with that

PLATE II



Pure culture of *R. bataticola*

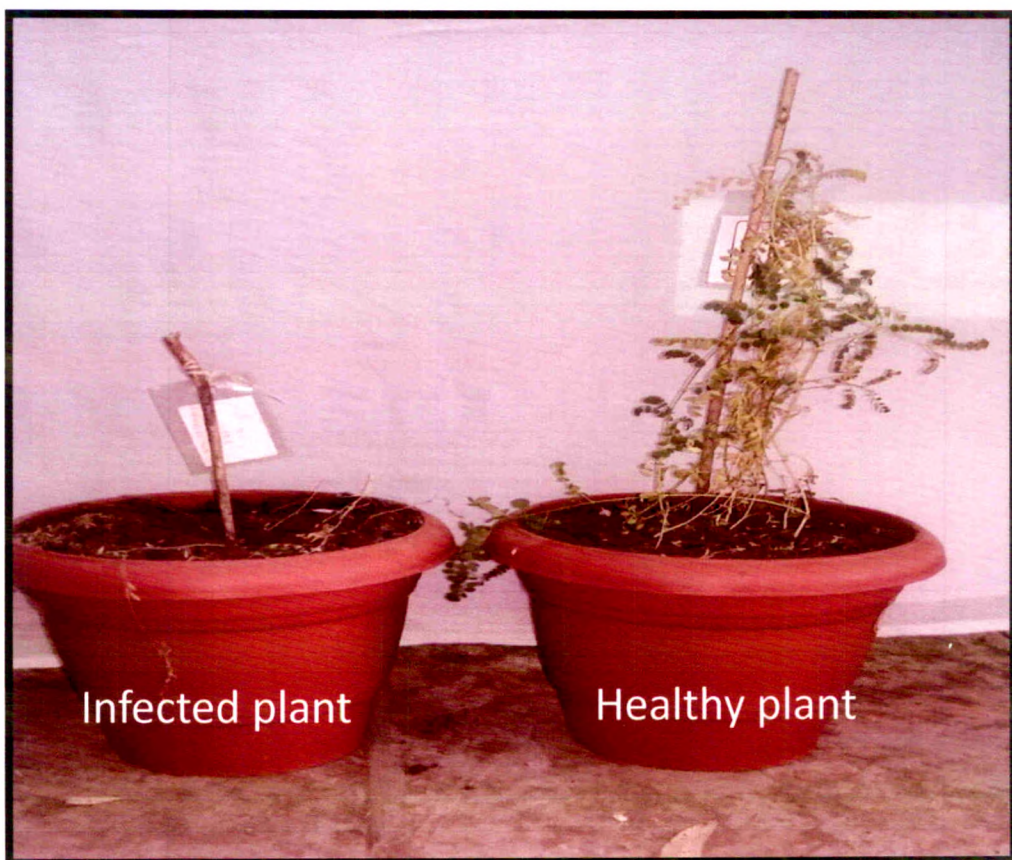


**Mass multiplication of *R bataticola* on
sand: maize medium**

PLATE III



Typical symptom of Dry root rot of chickpea



Pathogenicity of *R.bataticola*

of naturally infected chickpea plants (Plate-III) Earlier, Mitra (1931) proved pathogenicity of *R. bataticola* isolated from chickpea on potato. Bhatti *et al.* (1987), Beniwal *et al.* (1992) and Katariya *et al.* (2007) proved pathogenicity of *Rhizoctonia bataticola* on chickpea.

4.4. Reisolation

The fungus was re isolated on PDA from artificially inoculated plants showing typical dry root rot symptoms and the fungus growth was similar to original culture.

4.5. Morphological and cultural characters

4.5.1. Morphological characters

In the present investigation, the diversity in morphological and cultural characteristics viz., radial growth, colony colour and sclerotial features were studied from the growth on PDA after specific time.

4.5.1.1. Mycelium

The *Rhizoctonia bataticola* culture was inoculated and incubated on PDA medium at room temperature. After 24 hr, the mycelial growth started. Light microscopy studies revealed that in initial stage the colour of the mycelium was white, but after 48 hr gradually it changed and turned dark gray due to sclerotial formation. The mycelium was closely septate, blackish and thick walled, 6.86 μm (5.12 – 10.25 μm) in width and hyphal branches arised at right angles to the parent hyphae. After third day from incubation, the fungus produced sclerotia. On 6th day, sclerotia were dark gray coloured and irregular round in shape. In 8 days old culture the colony colour was blackish to dark black.

4.5.1.2. Sclerotial features :

The colour of the sclerotia was light black and finally turned black. Sclerotia varied in shape from irregular to oval or spherical, measuring 68.19 (48.71 $\times$ 87.17) μm in diameter.

These findings are in confirmity with those of earlier reports of viz., Bansal (1988) who reported mycelial character of *R. Bataticola* in groundnut, Short *et al.* (1978), Holliday (1980), Mihail (1992), Gupta and Kolte (1982) and Garg and Ashwinikumar (2012), who found that sclerotia were oval to spherical, dark brown to black and measured 62.5 μm to 100 μm in diameter in culture.

B) Cultural characters

4.5.1.3. Effect of different culture media on growth and sporulation of *Rhizoctonia bataticola*

Cultural characters of the test fungus were studied on nine different synthetic and semi-synthetic media. After seven days of inoculation the growth and colony characters of fungus on different media were noticed and the treatment differences in respect of colony diameter and growth characteristics observed. Colour change of different media was observed by using standard colour chart. Sporulation and spore count was recorded after 120 hrs. (Table 3) and depicted in Plate- 4 and Fig- 4.

4.5.1.4. Growth:

The results (Table 3) revealed that all the culture media tested encourages better growth except pline agar and Erlliots agar media. The mean colony diameter recorded in all the test media was ranged from 4.2cm (pline agar) to 9.00cm (PDA and host extract agar (chickpea)). However, significantly highest mean mycelia growth (9.00cm) was recorded on potato dextrose agar (PDA) and host extract agar chickpea. The second and thired best media found were Sourbounds agar (8.8cm) and Rhichard's agar (8.4cm) All of these four which statistiacly at par. This were followed by carrot agar (8.0cm) and Czapeks agar (7.8cm) were at par and were followed by oat meal aar (5.3cm), Elliots agar (4.4cm) which were at par. The minimum mean mycelial growth of the test pathogen was found on pline agar (4.2cm).

4.5.1.5 Colony characters

All the nine media tested exhibited varied colony characters. Potato dextrose agar (PDA) and host extract agar recorded circular colony ,thick growth of mycelium with dark grey colour of colony production of sclerotia. Submerge growth of mycelium with dull white colony without production of sclerotia noticed in oat meal agar and plane agar media. while carrot agar, Rhichards agar, Czapeks agar, Elliots agar exhibited circular colony of white coloured of mycelium with very poor production of sclerotia. The Saborounds agar produced regular colony with white to grayish mycelium with sclerotia.

4.5.1.6. Degree of sporulation

All the nine media tested reported a wide range of sporulation of test pathogen. However, potato dextrose agar (PDA) and host extract agar recorded excellent (++++) sporulation, Good sporulation (+++) was recorded Richards and Soboround's agar. Czapeks agar, carrot agar and Elliots agar exhibited fair (++) sporulation. where as no(-) sporulation was recorded on oat meal agar and pline agar.

Similar results were reported by EI Wakil *et al.* (1985), Sharma *et al.* (2004) and Khan *et al.* (2012), who found that potato dextrose agar, Host extract agar and Richard's agar supported good growth of *Rhizoctonia bataticola*.

Table 3. Effect of different cultural media on colony diameter, growth characters and sporulation of *Rhizoctonia bataticola*, causing dry root rot of chickpea.

Tr. No.	Medium	Average colony diameter (cm)* 10 days after inoculation	Colony characters	Degree of sporulation
1.	Potato dextrose agar	9.0	Colony circular, thick growth with entire margine, zonation present, colour of mycelium dark gray, sclerotia present.	++++
2.	Host extract agar (Chickpea)	9.0	Colony circular, thick growth of mycelium with dark grayish colour, regular and compact growth, sclerotia present.	++++
3.	Oat meal agar	5.3	Submerged growth of mycelium, sclerotia absent.	-
4.	Carrot agar	8.0	Colony circular, off white colour of mycelium, very poor sclerotia present.	++
5.	Pline agar	4.2	Submerged growth of mycelium, sclerotia absent.	-
6.	Richards agar	8.4	Colony circular, aerial snow white colour of mycelium, sclerotia present.	+++
7.	Czapeks agar	7.8	Colony circular, cottony white colour of mycelium, poor sclerotia present.	++
8.	Elliot's agar	4.4	Colony circular, submerged growth of mycelium, dull white colour of mycelium, sclerotia present.	++
9.	Saboround's agar	8.8	Regular colony, mycelium white at center grayish white at periphery, sclerotia present.	+++
	S.E. $\pm$	0.20		
	C.D. at 5%	0.60		

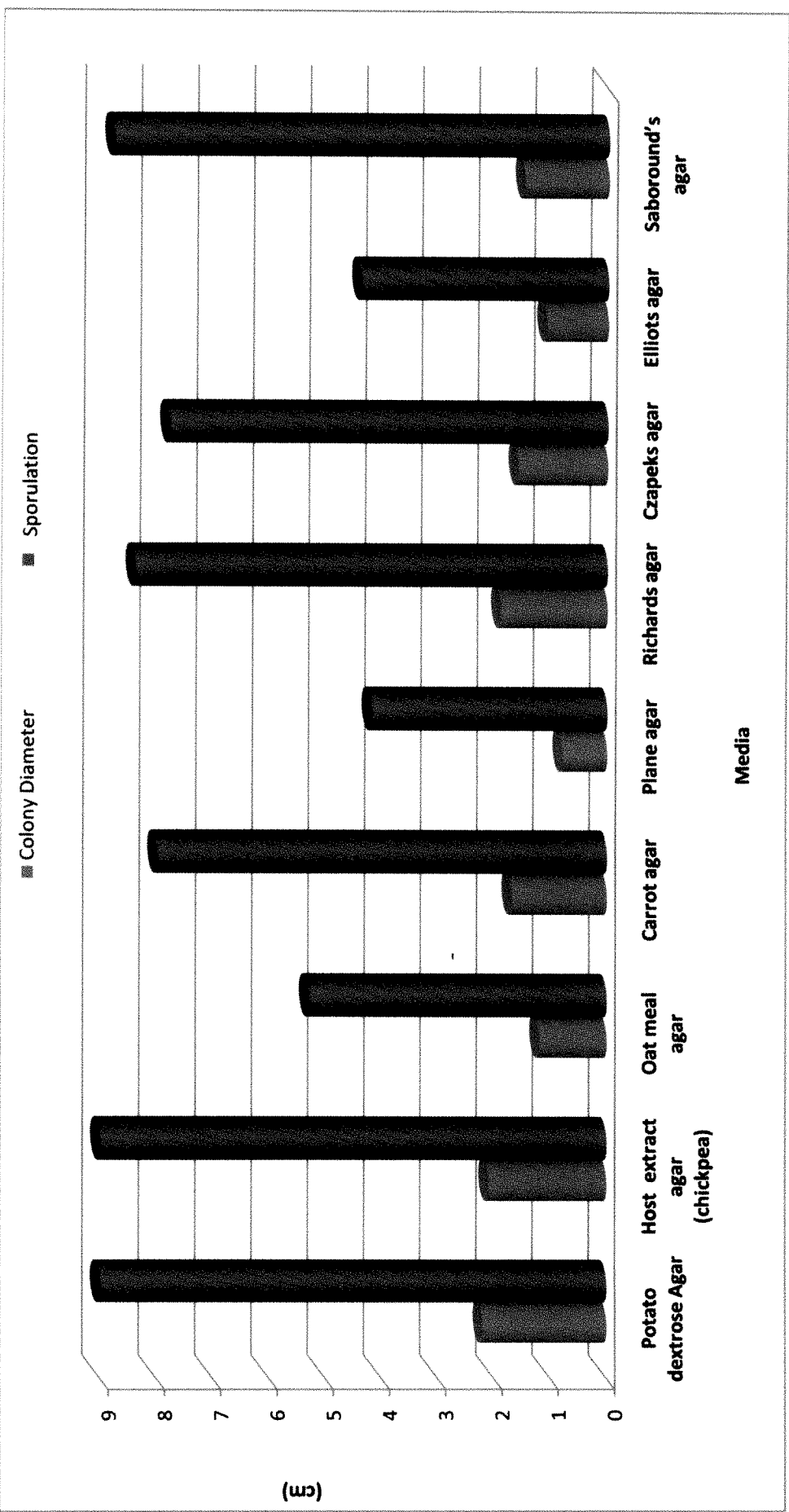
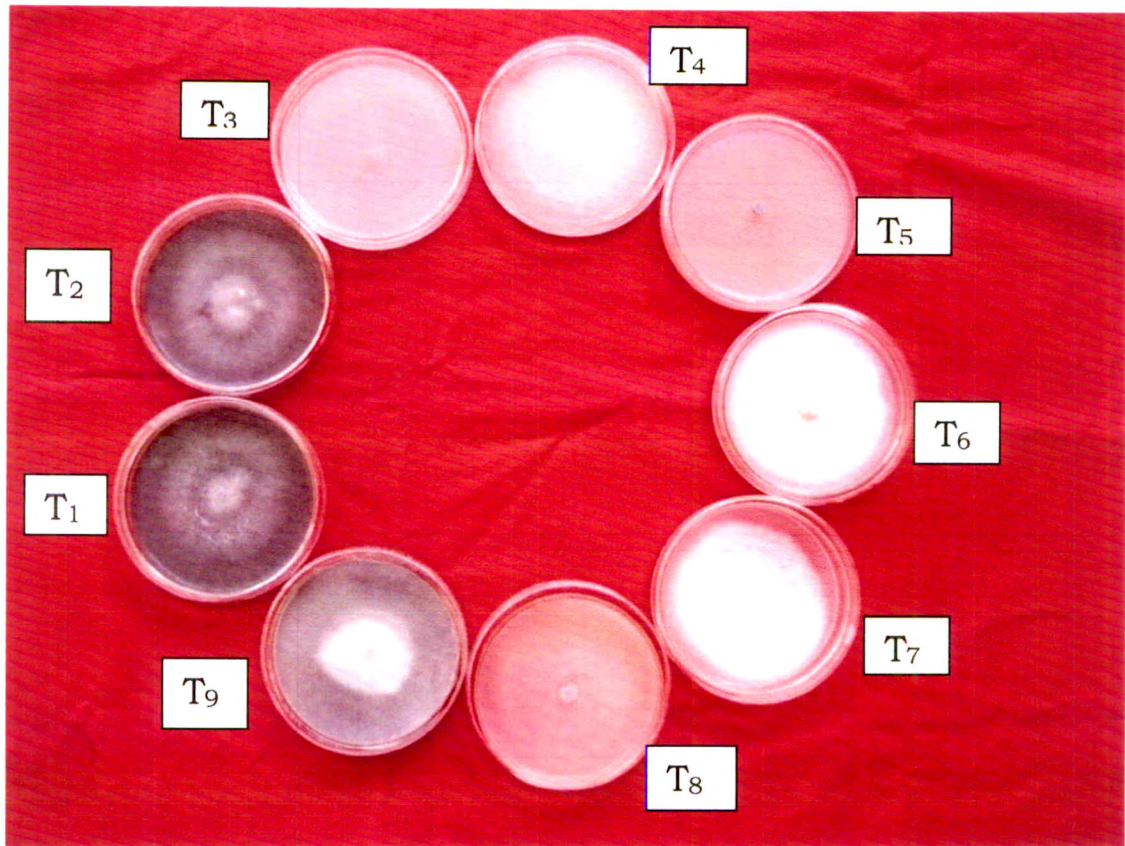


Fig. 4. Effect of different synthetic and semi-synthetic media on colony diameter and sporulation of *R. bataticola*.

PLATE IV



Effect of different cultural media on growth and sporulation of *Rhizoctonia bataticola*

T₁ Potato dextrose agar

T₂ Host extract agar

T₃ Oat meal agar

T₄ Carrot agar

T₅ Plane agar

T₆ Richard's agar

T₇ Czapek's agar

T₈ Elliot's agar

T₉ Saboround's agar

4.6.Disease management strategies

4.6.1.*In vitro* evaluation of fungicides

Results (Table 4 and Fig.5) revealed that fungicides evaluated *in vitro* against *R.bataticola* exhibited a wide range of mycelial growth of *R. bataticola* and was found to be decreased drastically and showed inhibition with increased concentration of fungicides tested.

4.6.1.1. Mycelial growth

At 500 ppm, radial mycelial growth of the test pathogen was ranged from 00.00 mm (carbendazim) to 37.94 mm (carboxin), as against 90.00 mm in untreated control. However, significantly least mycelium growth was recorded in carbendazim (0.00mm). This was followed by the fungicides viz., benomyl (07.22mm), Saff (08.19mm) and mancozeb (09.83mm) all these three were statistically at par. These were followed by thiram (16.07mm), propiconazole (26.95mm) and captan (30.12mm). Fungicide carboxin (37.94mm) was found comparatively less effective with maximum mycelial growth.

At 1000 ppm, all the 09 fungicides tested exhibited similar trend of mycelial growth as that of observed at 500 ppm, but growth was comparatively reduced and was ranged from 00.00 mm (carbendazim) to 35.58 mm (carboxin), as against 90.00 mm in untreated control. However, significantly least mycelial growth was recorded in carbendazim (00.00mm). This was followed by the fungicides viz., benomyl (05.08mm), Saff (6.09mm) and mancozeb (06.76mm) all these three were statistically at par and was followed by thiram (16.07mm), propiconazole (22.39mm) and captan (26.17mm). In fungicides, fungicide carboxine was found least effective with maximum colony diameter (35.58mm).

Table 4. *In vitro* efficacy of fungicides against mycelial growth and inhibition of *R.bataticola*.

Tr. No.	Treatments	Col. dia. *(mm) at conc.			Av. (mm)	% Inhibition*			Av. (%) inhibition
		500 ppm	1000 ppm	2000 ppm		500 ppm	1000 ppm	2000 ppm	
T ₁	Propiconazole (Tilt 25% EC)	26.95	22.39	13.06	20.07	70.06 (56.82)	75.12 (60.07)	78.58 (62.43)	74.58 (59.72)
T ₂	Carbendazim (Bavistin 50% WP)	00.00	00.00	00.00	00.00	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
T ₃	Captan (Captaf 75% WP)	30.12	26.17	19.19	25.16	66.52 (54.64)	70.91 (57.36)	78.83 (62.60)	72.08 (58.10)
T ₄	Carboxin (Vitavax 75 WP)	37.94	35.58	31.28	26.02	57.83 (49.50)	60.45 (51.03)	65.24 (53.87)	61.17 (51.45)
T ₅	Benomyl (Benlate 50 WP)	07.22	05.08	05.06	06.20	92.13 (73.70)	94.34 (76.23)	94.37 (76.27)	93.61 (75.35)
T ₆	Mancozeb (Indofil M-45 75% WP)	09.83	06.76	05.65	07.41	89.13 (70.74)	92.48 (74.08)	93.88 (75.67)	91.83 (73.39)
T ₇	Thiram (Thiram 75% WP)	16.07	16.07	11.94	14.69	82.13 (64.99)	82.13 (64.99)	86.72 (68.62)	83.66 (66.15)
T ₈	Carbendazim + mancozeb (Saff 75% WP)	08.19	06.09	05.00	06.69	90.89 (72.43)	93.25 (74.94)	94.59 (76.55)	92.91 (74.55)
T ₉	Control (Untreated)	90.00	90.00	90.00	90.00	000.00 (00.00)	000.00 (00.00)	000.00 (00.00)	000.00 (00.00)
	S.E.±	1.20	0.62	2.68	1.5	1.34	0.69	0.67	0.9
	C.D. (P=0.01)	3.27	1.68	7.29	4.08	3.64	1.87	1.82	2.44

*-Mean of three replications, Col. = Colony, Dia. = Diameter, Conc. = Concentration, Av. =Average, Figures in parentheeis are arc sine transformed value.

At 2000 ppm, all the fungicides tested exhibited some what similar trend of mycelial growth as that of observed at 500 ppm and 1000 ppm and it was ranged from 00.00 mm (carbendazim) to 31.28 mm (carboxin), as against 90.00 mm untreated control. However, with the fungicide carbendazim significantly least

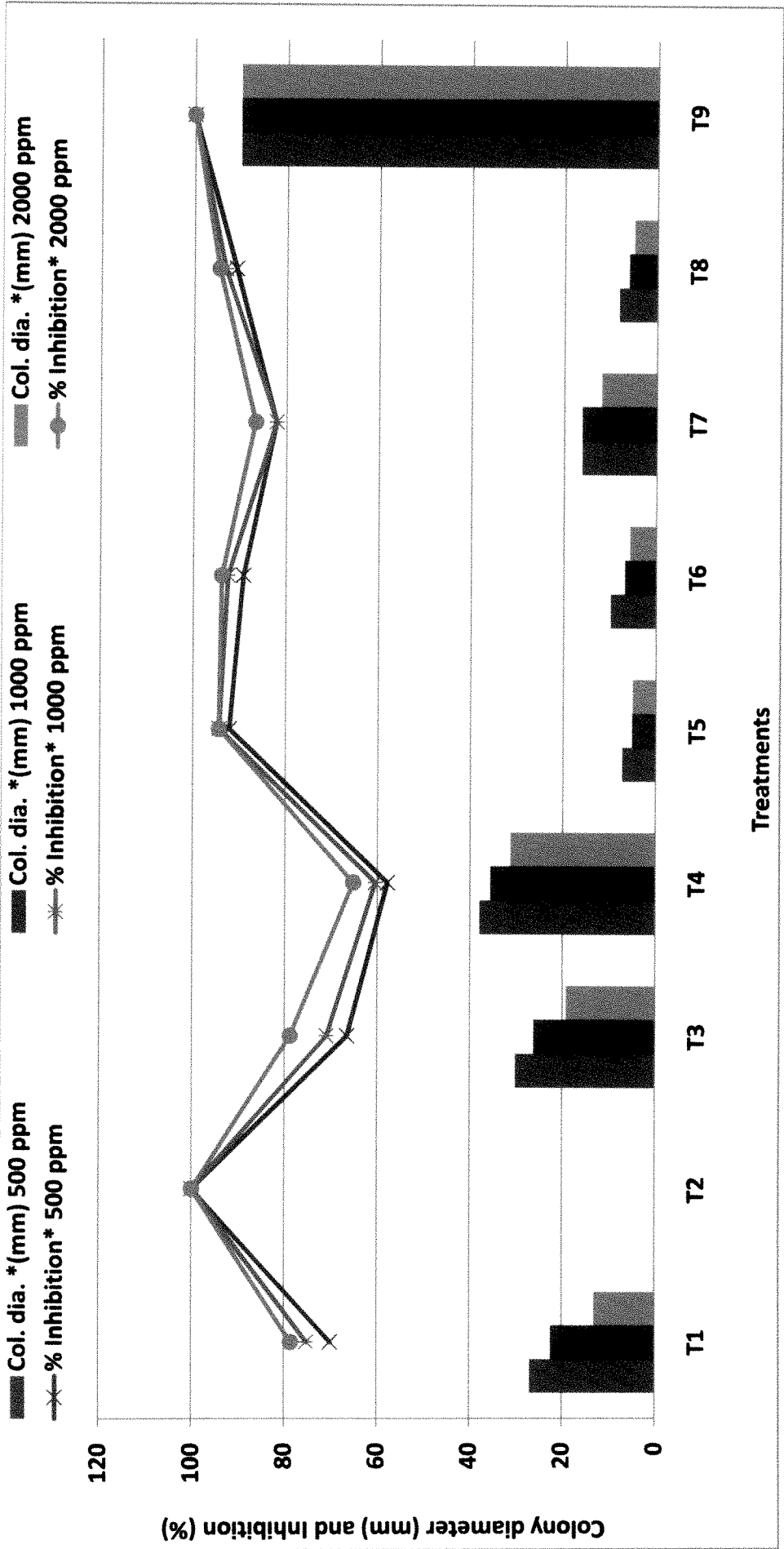
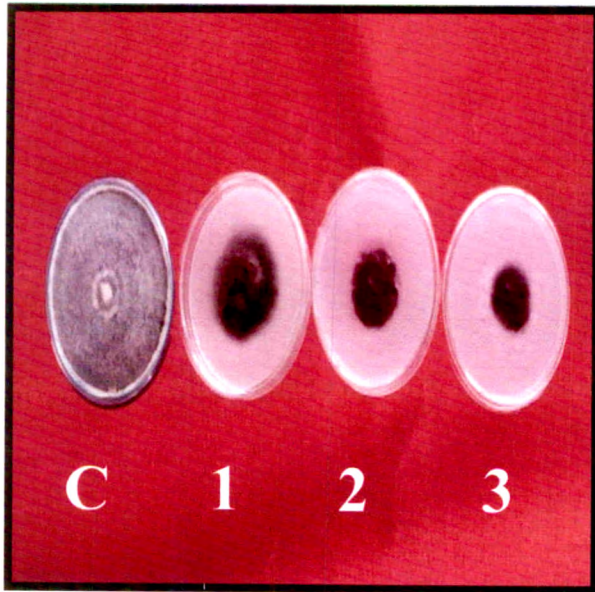
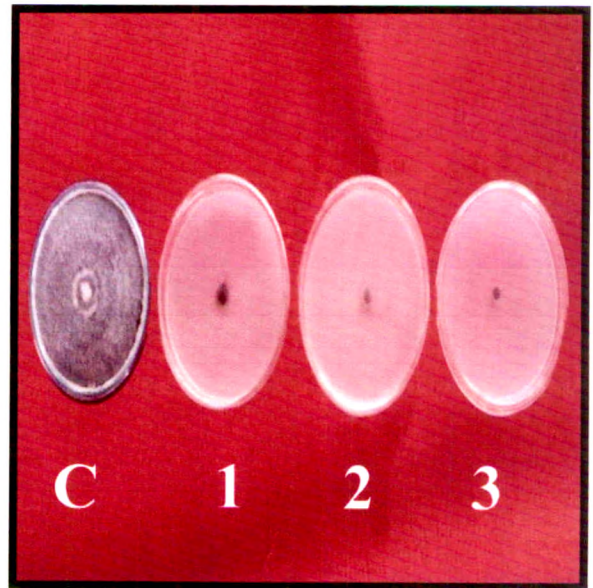


Fig. 5. *In vitro* efficacy of fungicides against mycelial growth and inhibition of *R. bataticola*.

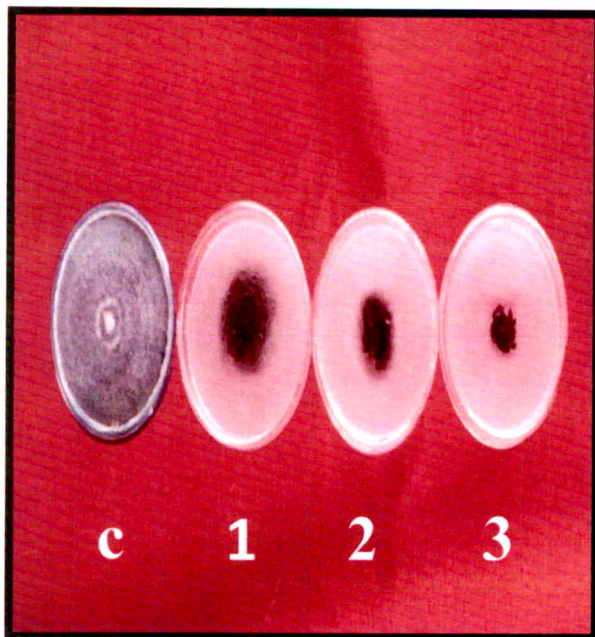
PLATE V



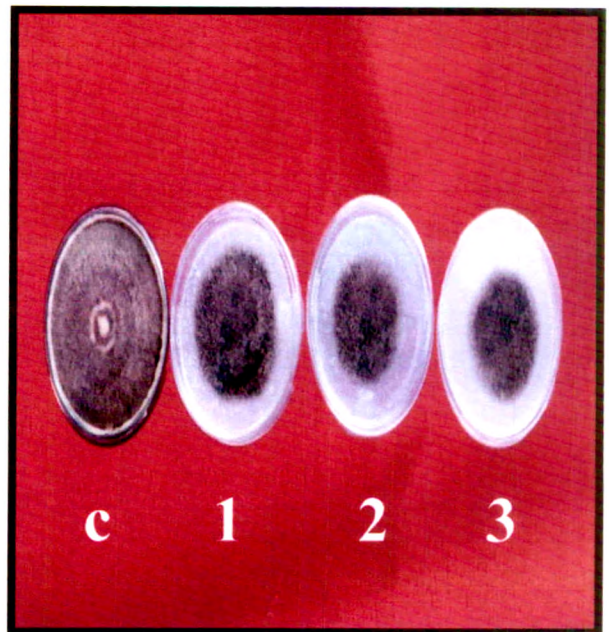
Propiconazole



Carbendazim



Captan

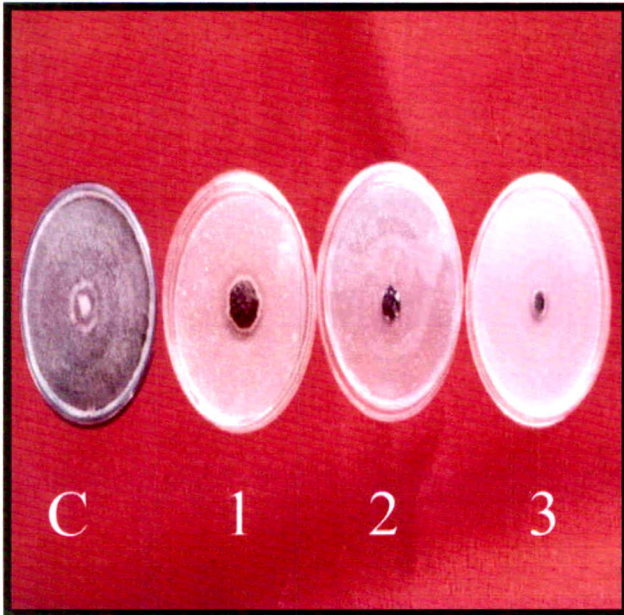


Carboxin

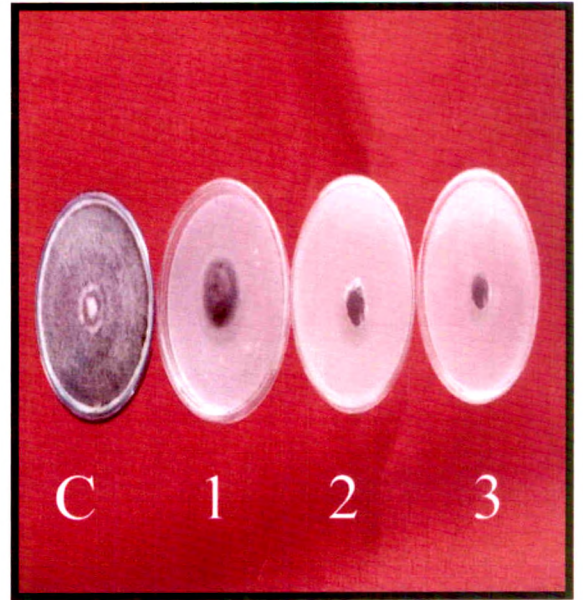
Effect of fungicide on mycelial growth of *R. bataticola*

C= Control, 1=500ppm, 2=1000ppm, 3=2000ppm

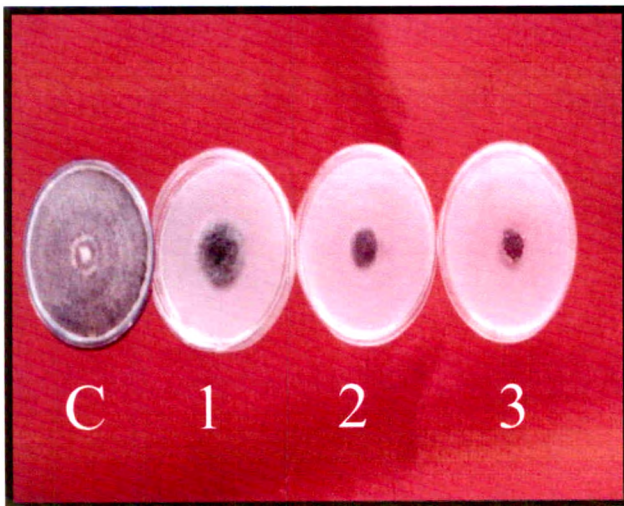
PLATE VI



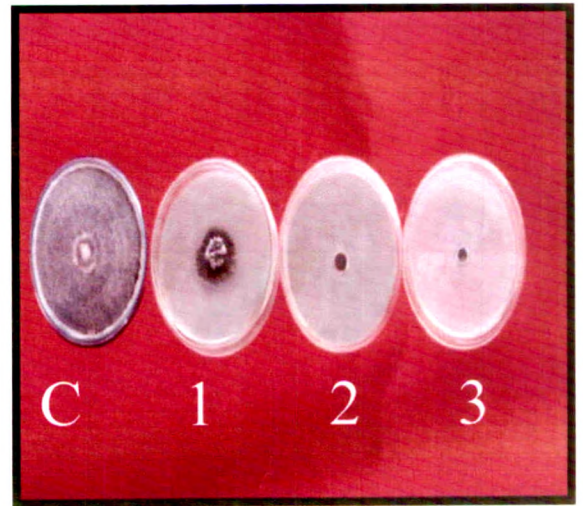
Benomyl



Mancozeb



Thiram



Carbendazim+Mancozeb

Effect of fungicide on mycelial growth of *R. bataticola*

C= Control, 1=500ppm, 2=1000ppm, 3=2000ppm

mycelium growth (00.00mm) was observed. This was followed by the fungicides viz., Saff (05.00mm), benomyl (05.06mm), mancozeb (5.65mm) and thiram which at par. Followed by other fungicides propiconazole (13.06mm) and captan (26.17mm), while the fungicide carboxin was recorded comparatively less effective with maximum mycelia growth of (31.28mm).

Average radial mycelial growth recorded in all the 09 fungicides tested was ranged from 00.00 mm (carbendazim) to 26.02 mm (carboxin), as against 90.00 mm in untreated control. However, the least average mycelia growth was recorded with carbendazim (00.00mm). This was followed by the fungicides viz., benomyl (06.20mm), Saff (06.69mm) and mancozeb (07.41mm) all these three were at par and was followed by thiram (14.69mm), propiconazole (20.07mm) and captan (25.16mm). the highest average colony diameter was recorded in carboxin (26.02mm).

4.6.1.2. Mycelial growth inhibition

Results (Table 4, Fig.5) revealed that all the fungicides tested (@ 500, 1000 and 2000 ppm each) significantly inhibited mycelial growth of *R. bataticola* over untreated control (00.00%). Further, the percentage mycelial growth inhibition of test pathogen was increased with increase in concentrations of the fungicides tested (PLATE 5 and 6).

At 500 ppm, percentage mycelial growth inhibition of the test pathogen was ranged from 57.83 (carboxin) to 100.00 per cent (carbendazim). However, significantly higher mycelia inhibition was recorded with fungicide carbendazim (100.00%). The second, third and fourth best fungicides found were benomyl (92.13%) and Saff (90.89%) and mancozeb (89.13%) which were at par followed by thiram (82.13%), propiconazole (70.06%), captan (66.52%). The fungicide carboxin was found less effective with 57.83 per cent inhibition of the test pathogen.

At 1000 ppm, mycelial growth inhibition was increased compared to 500 ppm and similar trends of mycelial growth inhibition were observed. It was ranged from 60.45 (carboxin) to 100.00 (carbendazim) per cent. However, significantly highest mycelia inhibition was found with fungicide carbendazim (100.00%). The second, third best fungicides, viz., benomyl (94.34%), Saff (93.25%) both were at par. These were followed by fungicides viz., mancozeb (92.48%), thiram (82.13%), propiconazole (75.12 %), and captan (70.91%). The fungicide carboxin was found less effective with 60.45mm inhibition of the test pathogen.

At 2000 ppm, all the 09 fungicides tested exhibited comparatively increased mycelial growth inhibition than that of at 500 and 1000 ppm and it was ranged from 65.24 (carboxin) to 100 (carbendazim) per cent. However, significantly highest mycelia inhibition was recorded with fungicide carbendazim (100.00%). This was followed by the fungicides, viz., saff (94.59%), benomyl (94.37%) and mancozeb (93.88) were at par followed by thiram (86.72%). The fungicides captan (78.83%), propiconazole (78.58%) were at par the fungicide carboxin was reported comparatively less effective with growth inhibition of 65.24%.

Average mycelial growth inhibition recorded in the test fungicides was ranged from 61.17 (carboxin) to 100.00 (carbendazim) per cent. However, significantly by highest average mycelial growth inhibition was recorded in carbendazim (100.00%). The second and third best fungicides found were benomyl (93.61%) and Saff (92.91%), both were at par. These were followed by the fungicides viz., mancozeb (91.83%), thiram (83.66%), propiconazole (74.58%), captan (72.08%). The fungicide carboxin was comparatively less effective with minimum growth inhibition of 61.17 per cent. Thus, all the fungicides tested were found fungistatic against *R. bataticola* and significantly inhibited its mycelial growth over untreated control. However, fungicides found most effective in the order of merit were carbendazim, benomyl, Saff, mancozeb, thiram, propiconazole and captan.

Similar fungistatic effects of the test fungicides against *R. bataticola* infecting chickpea and many other crops were reported earlier by several workers, Goyal and Mehrotra 1981, Taneja and Grover 1982, Sarwar and Gopal Raju 1985, Ramadoss and Shivprakasham 1987, Singh and Sindan 1988, Peshney *et al.* 1992, Prajapati *et al.* 2004, Sandipan *et al.* 2007, Date 2015.

4.6.2. *In vitro* evaluation of bioagents

The results obtained on mycelial growth and inhibition of *R. bataticola* with eight antagonists are presented in Table 5 and depicted in Fig.6 and PLATE 7. Results revealed that all the bioagents evaluated exhibited fungistatic / antifungal activity against *R. bataticola* and significantly inhibited its growth over untreated control.

Table 5. *In vitro* efficacy of bioagents against mycelial growth and inhibition of *R. bataticola*.

Tr. No.	Treatments	Colony diameter* (mm)	% Inhibition
T ₁	<i>Trichoderma viride</i>	10.24	88.62 (70.28)
T ₂	<i>T. harzianum</i>	30.30	66.33 (54.52)
T ₃	<i>T.hamatum</i>	34.51	61.65 (51.73)
T ₄	<i>T. longibrachiatum</i>	38.02	57.75 (49.45)
T ₅	<i>T. koningii</i>	39.12	56.53 (48.75)
T ₆	<i>T. virens</i>	38.44	47.28 (43.44)
T ₇	<i>Aspergillus niger</i>	40.38	55.13 (47.94)
T ₈	Control	90.00	00.00 (00.00)
	S.E. $\pm$	0.63	
	C.D.@ 1%	1.86	

*-Mean of three replications, Figures in parentheses are arc sine transformed values.

Among the bioagents/antagonists tested, *T. viride* was found most effective and recorded significantly least colony diameter (10.24mm) with highest per cent inhibition (88.62%) of the test pathogen. The second and third best antagonists found were *T. harzianum* and *T.hamatum* which recorded colony diameter of 30.30 mm and 34.51 mm and per cent inhibition of 66.33 and 61.65 per cent, respectively. These were followed by *T.longibrachiatum* (38.02mm and 57.75%), *T.koninigi* (39.12mm and 56.53%) and *T. virens* (38.44mm and 47.28%) of colony diameter and its per cent inhibition, respectively. *A. niger* was found comparatively

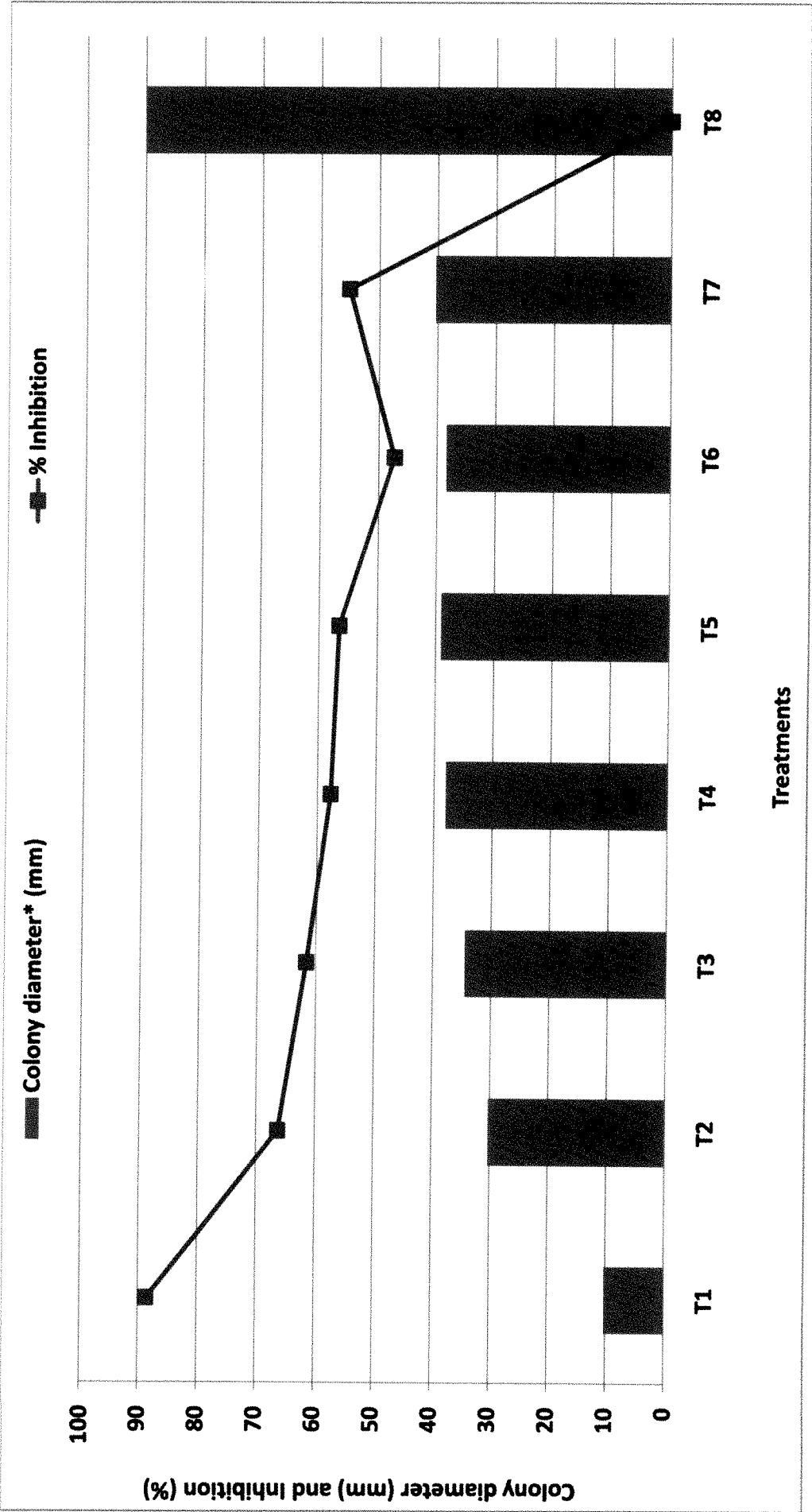
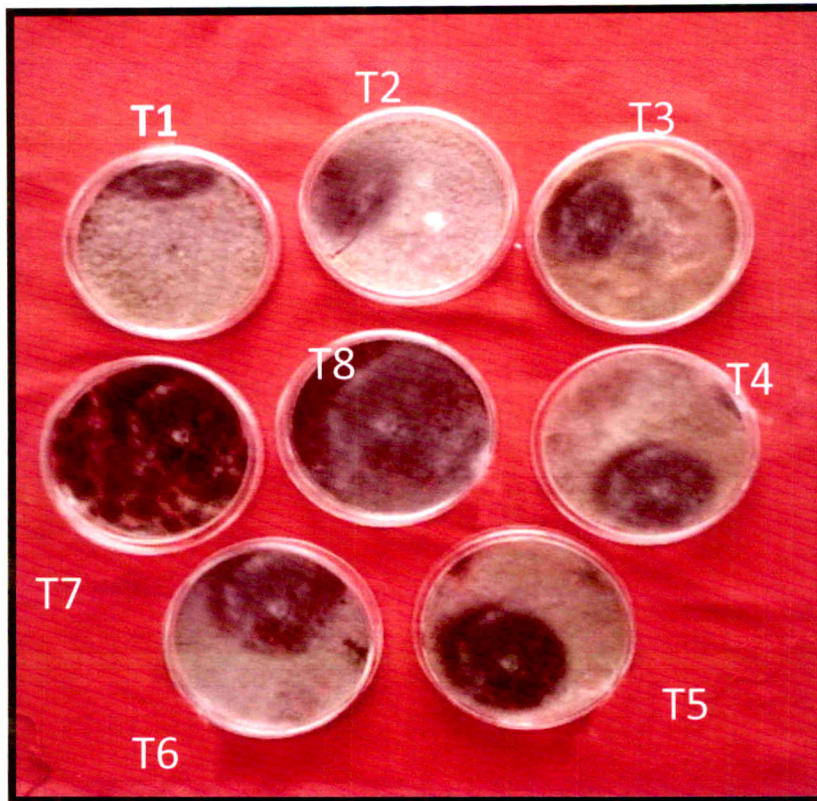


Fig. 6. *In vitro* efficacy of bioagents against mycelial growth and inhibition of *R. bataticola*

PLATE VII



Effect of bioagents on *R. bataticola*

T1= T. viride

T2=T. harzianum

T3=T.hamatum

T4=T.longibrachiatum

T5=T.konigii

T6=T.virens

T7=Aspergillus niger

T8=Control

less effective with maximum colony diameter (40.38mm) and minimum per cent mycelial inhibition (55.13%).

Results of the present study on antifungal activity of *T. viride*, *T. harzianum*, *T. hamatum*, *T. longibrachiatum*, *T. virens*, *T. koningii* and *Aspergillus niger* against *R. bataticola* are in conformity with those reported earlier by several workers Malathi and Doraisamy 2003 and Joytsana *et al* 2008.

T. viride showed best performance against the pathogen followed by *T. harzianum*, *T. hamatum*, *T. longibrachiatum*, *A. niger*, which also checked the fungus growth to some extent similar finding were also reported by Singh *et al.* (2006). Pan (2009) reported the antagonistic potential of *Trichoderma* isolates through production of volatile and non volatile substance against *M. phaseolina*. Choudhary (2010) tested bioagents viz., *T. viride*, *T. harzianum*, *A. vesicolor* for the control of dry root rot in mungbean. In dual culture, *T. viride*, *T. harzianum* and *A. vesicolor* were effective in inhibiting the growth of *M. phaseolina* to an extent of 61-65 per cent.

4.6.3. Effect of organic amendments

A total of seven amendments were evaluated as pre-sowing application against *R. bataticola* (sick soil), sowing susceptible chickpea cv.JG-62 in pot culture under screen house conditions. The results obtained on percentage seed germination, pre-emergence seed rot (PESR) and post-emergence seedling mortality (PESM) are presented in Table 6 and depicted in PLATE 8 and Fig.7.

4.6.3.1. Effect on seed germination

Results (Table 6, PLATE 8 and Fig.7) revealed that all the test amendments significantly improved the percentage seed germination over untreated control (sick soil) and it was ranged from 40.46 to 83.51 per cent, as against 45.00 per cent in untreated control (sick soil). However, neem seed cake was found most effective with significantly highest seed germination (83.51%). The next best treatment was FYM (73.39%), both were statistically at par, those were followed Vermicompost (66.71%), Mustard cake (60.00%) and castor cake (56.69%). However, castor cake and poultry manure were found least effective with comparatively minimum seed germination of 56.67 and 46.46 per cent, respectively.

4.6.3.2. Effect on pre- and post-emergence mortalities

Results (Table 6, Fig. 7) revealed that all the test amendments significantly influenced both pre-emergence seed rot (PESR) and post emergence seedling mortality (PESM), caused by *R. bataticola* in chickpea cv.JG-62. The pre-emergence seed rot (PESR) recorded in all the test amendment were ranged from 16.60 to 46.60 per cent, as against 00.00 per cent in untreated control (sick soil). However, significantly least pre-emergence seed rot was recorded in neem seed cake (16.60%). This was followed by the amendments viz., FYM (26.71%), vermicompost (33.52%), mustard cake (40.00%), castor cake (43.45%) and poultry manure (46.60%). However, castor cake and poultry manure were found least effective with comparatively maximum PESR of 33.52 and 46.60 per cent, respectively.

Similar trend in respect of the post- emergence seedling mortality (PESM) was also observed and it was ranged from 24.40 to 48.84 per cent, as against 80.00 per cent in untreated control (sick soil). Of the amendments tested, neem seed cake was found most effective with significantly least post-emergence seedling mortality 24.40 per cent. It was followed by the amendments viz., FYM (31.54%), vermicompost (39.67%), mustard cake (44.33%), castor cake (47.69%) and poultry manure (48.84%). However, castor cake and poultry manure were found least effective with comparatively higher PESM of 47.67 and 48.84 per cent, respectively.

The average mortality recorded in all the test amendments were ranged from 20.05 to 47.72 per cent, as against 40.00 per cent in untreated control (sick soil). However, significantly least average mortality was recorded with Neem seed cake (20.05%), followed by FYM (29.12%), this was followed by viz., vermicompost (36.59%), mustard cake (42.16%), castor cake (45.59%) and poultry manure (47.72%). Whereas, poultry manure and castor cake were found least effective with comparatively higher average mortality of 47.72 and 45.59 per cent, respectively.

Table 6. Efficacy of organic amendments soil application (pre sowing) against *R. bataticola* causing dry root rot in chickpea JG-62 (pot culture).

Tr. No.	Treatments	Germination (%)	Rot/ mortality (%)*		Average mortality (%)	Reduction over control (%)		Average reduction (%)
			PESR	PESM		PESR	PESM	
T ₁	FYM	73.39 (47.20)	26.71 (15.49)	31.54 (18.38)	29.12 (16.93)	62.07 (38.36)	60.56 (37.27)	61.31 (37.81)
T ₂	Castor cake	56.69 (34.53)	43.50 (25.75)	47.69 (28.48)	45.59 (27.11)	38.13 (22.41)	40.37 (23.81)	39.25 (23.11)
T ₃	Vermicompost	66.71 (41.84)	33.52 (19.58)	39.67 (23.37)	36.59 (21.47)	52.10 (31.39)	50.37 (30.24)	51.23 (51.23)
T ₄	Poultry manure	40.46 (24.28)	46.60 (27.77)	48.84 (29.23)	47.72 (28.05)	33.41 (19.51)	38.94 (22.91)	36.17 (21.21)
T ₅	Mustard cake	60.00 (36.86)	40.00 (23.57)	44.33 (26.31)	42.16 (24.94)	42.68 (25.26)	44.58 (26.47)	43.63 (25.86)
T ₆	Neem seed cake	83.51 (56.62)	16.60 (09.55)	24.04 (14.12)	20.05 (11.83)	76.27 (49.70)	69.49 (44.02)	72.88 (46.86)
T ₇	Control (untreated)	45 (26.88)	00.00 (00.00)	80.00 (53.12)	40.00 (26.56)	00.00 (00.00)	00.00 (00.00)	00.00 (00.00)
	S.E. $\pm$	4.71	0.21	0.22	0.21	0.35	0.28	0.31
	C.D. (P=0.05)	13.91	1.27	0.87	1.07	1.42	1.11	1.26

*-Mean of three replications, PESR- Pre emergence seed rot, PESM- Post emergence seedling mortality, Figures in parentheses are arc sine transformed values.

4.6.3.3.Per cent reduction over control:

Result presented in (Table 6) revealed that all the treatments were effective against *R. bataticola* and significantly reduced both pre emergence(PESM) seed rot as well as post emergence mortalities (PESM) in chickpea cv. JG-62.

The percentage reduction in PESR recorded in all the treatments was ranged from 33.41 to 76.27 per cent over untreated control (00.00%). However, the neem seed cake was most effective and recorded significantly highest reduction (76.27%) in PESR. This was followed by treatments FYM (62.07%), vermicompost (52.10%), mustard cake (42.68%), castor cake (38.13%) and poultry manure (33.41%).

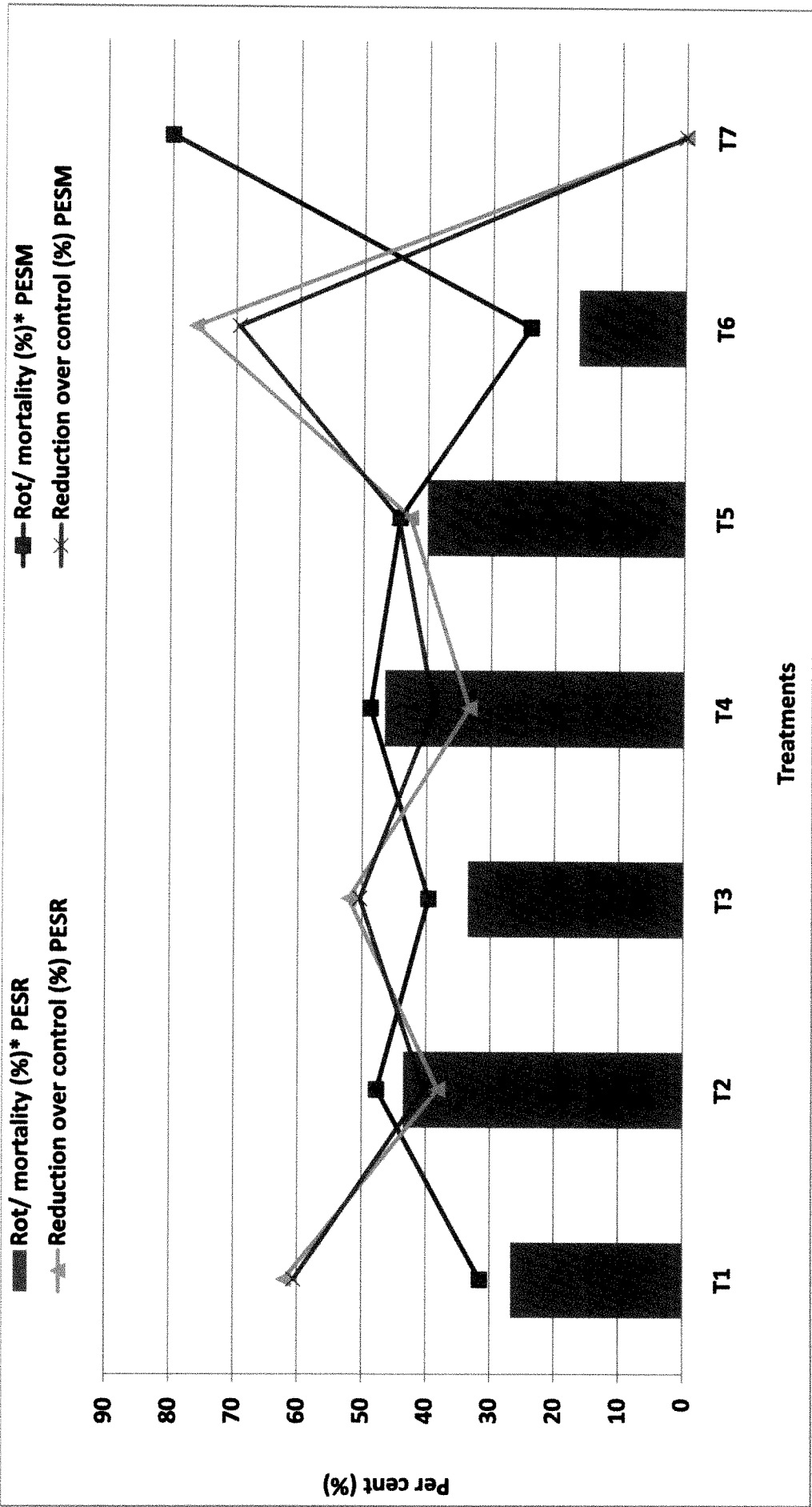


Fig. 7. Efficacy of organic amendments soil application (pre sowing) against *R. bataticola*

PLATE VIII



Efficacy of organic amendments on *R. bataticola*

T1=FYM

T2=Castor cake

T3=Vermicompost

T4=Poultry manure

T5=Mustard cake

T6=Neem seed cake

T7=Control

The per cent reduction in PESR recorded in all the treatments was ranged from 38.94 to 69.49 per cent over untreated control (00.00%). However, the organic amendments neem seed cake was found most effective and recorded significantly highest reduction (69.49%) in PESM. The second and thired best treatments of organic amendments found were FYM (60.56%)(50.37%). This was followed by mustard cake (44.58%), castor cake (40.37%), poultry manure (38.94%).

4.6.3.4. Average reduction

The data from (Table 6) reavealed that,the average per cent mortality reduction (PESR and PESM) was recorded with all the test amendments and it was ranged from 00.00 (poultry manure) to 72.88 (neem seed cake) per cent over untreated control. However, significantly highest average mortality reduction (72.88%) was recorded with neem seed cake. This was followed by the amendments viz., FYM (61.31%), vermicompost (51.23%),musatrd cake (43.63%) and castor cake (43.99%). Less than 40.00 per cent average mortality reduction was recorded with Poultry manure (39.29%), Vermicompost (36.11%) and Castor cake (39.25%); the least effective amendment was Poultry manure (36.17%).

Results of the present study obtained on the efficacy of the soil amendments against *Rhizoctonia bataticola* are in conformity with those reported earlier by several workers. Organic amendments viz., FYM, castor cake, poultry manure, Mustard cake, vermicompost and neem seed cake were reported earlier by several workers Muthuswami and Mirippan, 1991; Hundekar *et al.*, 1998; Savarna *et al.*, 2004; Suriachandraselvan *et al.*, 2005; Sudha *et al.*, 2008; Jaiman *et al.*, 2009.

4.7. Integrated management of *Rhizoctonia bataticola* (pot culture)

Those fungicides, bioagents and organic amendments found most effective against *R. bataticola* under *in vitro* studies were integrated and evaluated for the management of *R. bataticola* under green house condition (pot culture).

The pot culture experiment was conducted during *Rabi* season of 2015-16 to evaluate the bioefficacy of three fungicides, three bio agents and two organic amendments by seed treatment and soil application method in Pot culture with susceptible Chickpea cultivar JG-62 against *R. bataticola*.

Result (Table-7) revealed that all the treatments were found effective against the test pathogen and significantly enhanced the seed germination and

Table 7. Integrated management of *Rhizoctonia bataicola* (Pot culture)

Sr.No	Treatment	Concentration	Method of application	Germination %	% Disease incidence	%Reduction over control
1	Carbendazim 50WP	1%	Seed treatment	93.33 (76.23)	31.15 (33.92)	71.09 (57.47)
2	Benomyl 50WP	1%	Seed treatment	83.33 (66.14)	26.42 (30.82)	87.35 (65.91)
3	(SAFF)75WP Carbendazim+mancozeb	2%	Seed treatment	83.33 (66.14)	34.81 (36.13)	69.04 (56.41)
4	<i>T.viride</i> 5×10 ⁶ CFU/GM	10%	Seed treatment	90.00 (73.65)	11.56 (19.85)	65.91 (54.27)
5	<i>T.harzianum</i> 5×10 ⁶ CFU/GM	10%	Seed treatment	80.00 (63.43)	28.29 (32.06)	61.91 (51.89)
6	<i>T.hamatum</i> 5×10 ⁶ CFU/GM	10%	Seed treatment	78.00 (62.02)	45.89 (43.64)	49.75 (44.85)
7	Neem seed cake	50%	Soil application	86.67 (68.86)	20.77 (26.66)	77.27 (61.52)
8	FYM	50%	Soil application	70.00 (56.79)	50.34 (45.19)	44.91 (42.07)
9	Control	-		63.33 (52.78)	91.39 (75.10)	00.00
	SE			3.02	2.70	
	CD at 5%			9.50	8.70	

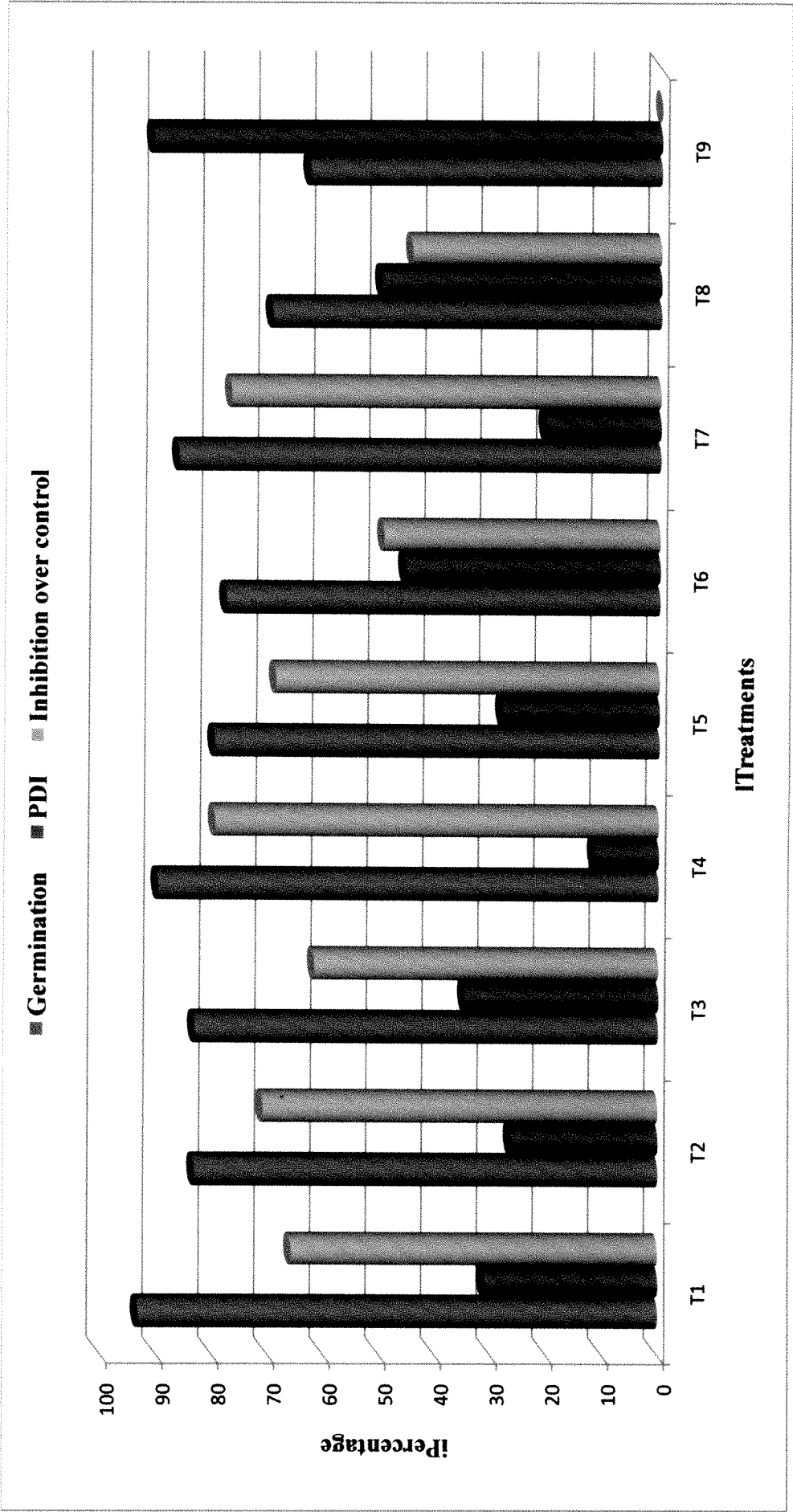


Fig. 8. Evaluation of fungicides, bioagents and organic amendments against *Rhizoctonia bataticola* in chickpea cultivar JG-62 (pot culture).

PLATE IX



Efficacy of fungicides, Bioagents and Organic amendments on *R. bataticola*

- T1=FYM**
- T2=Castor cake**
- T3=Carbendazim+Mancozeb**
- T4=*T. viride***
- T5=*T. harzianum***
- T6=*T. hamatum***
- T7=neem seed cake**
- T8=FYM**
- T9=Control**

reduced the percent disease incidence in Chickpea over untreated control. It was ranged from 70.00-93.33 per cent (Plate-10). Among all treatment, maximum per cent germination was recorded with seed treatment with Carbendazim (93.33%). It was followed by seed treatment *T. viride* (90.00 %), and soil application of Neem seed cake (86.67 %). These treatment were at par with each other, it was followed by Benomyl (83.33 %), saff (83.33 %), *T. harzianum* (80.00%), *T. hamatum* (78.00 per cent), FYM (70.00 %) as compared to untreated control (66.33%)

The result (Table 7) indicated that all treatments significantly influenced the disease incidence was recorded to control. The least disease incidence was recorded in seed treatment with *T. viride* (11.56%). The next best treatment was application neem seed cake (20.77%) at par with each other followed by benomyl (26.42% and *T. harzianum* (28.29%), seed treatment with carbendazim (31.15%), saff (34.81%), *T. hamatum* (45.89%) and FYM (50.34%). The treatment viz., carbendazim, saff, FYM in order of merit. The treatment having the disease incidence of viz., carbendazim, saff, FYM, over the control which exhibited (91.39%).

Per cent reduction of Dry root rot disease incidence over control was maximum with seed treatment, *T. viride* (79.83%), it was followed by soil application of neem seed cake with *T. harzianum* (69.04%), in carbendazim (65.91%), saff (61.91%), *T. hamatum* (49.78%), the least disease % was recorded in the treatment of application recorded (44.86%).

Results of the present study obtained on the integrated management of (*Rhizoctonia bataticola*) with fungicides bioagents and organic amendments are in conformity with those reported earlier by several workers Bhattacharya *et al.* (1956), Datar (1999), Kumar *et al.* (2007), Savarna *et al.* (2004); and Suriachandraselvan *et al.* (2005).

8. Screening of chickpea varieties against dry root rot

Twenty five varieties were evaluated in pot culture against dry root rot of chickpea in artificially sick soil in search of genetic donors and data is represented in Table 8.

Table -8 Evaluation of chickpea varieties against root rot

Sr. No.	Genotypes	Incidence of root rot (%)	Disease reaction
1	PG0302-10	12.65	MR
2	Digvijay	17.53	MR
3	AKG 1103	50.00	S
4	SAKI 9516	08.09	R
5	BCP 26	04.28	R
6	Vishal	16.57	MR
7	BDNG 13-1	09.02	R
8	BDNG 2013-2	54.56	HS
9	BDNG 20B-1	06.08	R
10	JAKI 9218	13.06	MR
11	BDNG797	24.18	MS
12	BDNG 9-3	21.86	MS
13	Vijay	39.96	S
14	PG -9801	67.23	HS
15	BCP -49	06.64	R
16	BCP -21	06.08	R
17	JG -315	06.67	R
18	JG – 62	100	HS
19	PG -0014-6-3	67.23	HS
20	PG – 0102-11-2	68.89	HS
21	ICCV – 5313	24.93	MS
22	Rajas	06.66	R
23	PG – 4333	77.07	HS
24	PG –O515	10	R
25	PG-9758	98.02	HS

Based on disease rating, all the varieties of chickpea evaluated were categorized as Immune (I), Resistant (R), Moderately Resistant (MR), Susceptible (S) and very susceptible (VS). The data recorded in Table 9 showed that

PLATE X



Varietal screening against *R. bataticola*

none of entry was free from dry root rot. It is evident from the Table 8 that out of 25 varieties, 09 varieties viz SAKI-9516, BCP-26, BDNG20B-1, JG-315 (R-ch), Rajas, PG-0515, BCP-21, BDNG 13-1, BCP-49 were found resistant. The 06 varieties viz PG0302-10, Digvijay, Vishal, JAKI 9218, ICCV-5313, BDNG-9-3 were found moderately resistant. The 01 varieties i.e. ICCV-5313 was found moderately susceptible. The two varieties viz AKG 1103, Vijay were found susceptible. The 07 lines viz BDNG 2013-2, PG9801, JG-62(S-W.Ch), PG0014-6-3, PG-0102-11-2, PG-9758, PG4333 were found highly susceptible.

The screening of varieties showed that there is variation in resistance level of germplasm lines. The present results are confirmly with earlier reports. Differential responses of chickpea genotypes to dry root rot have been reported by Singh and Mehrotra (1982), Nene *et al.* (1989), Reddy *et al.* (1991), Bekele *et al.* (1992), Bhatt (1993), Gangwar *et al.* (2002), Pande *et al.* (2004), Katariya and Gaur (2007) and Jaylakshmi *et al.* (2008).

Table 9. Categorization of chickpea varieties on the basis of their reaction dry root rot.

Disease Reaction	Rang	Genotypes
R	6.08-10	SAKI – 9516, BCP – 26, BDNG – 13-1, BDNG – 20B-1, BCP – 49, BCP – 21, JG – 315, PG – 0515, Rajas
MR	12.65-17.53	PG – 0302-10, Digvijay, Vishal, JAKI – 9218, BDNG – 793, BDNG – 9-3
MS	21.86-24.93	ICCV – 5313
S	39.96-50	AKG – 1103, Vijay
HS	54.56-100	BDNG-2013-2, PG-9801, JG – 62, PG – 0014-6-3, PG-0102-11-2, PG-4333, PG – 9758



Summary and Conclusions



CHAPTER V

SUMMARY AND CONCLUTION

The cultivated chickpea (*Cicer arietinum* L) was one of the first grain legume to be domesticated in the old world. The chickpea crop is attacked by 172 pathogen, viz., 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma from all over the world (Nene *et al.* 1996). Among all only a few of them have the potential to devastate the crops. Some of the serious diseases in order of their importance are wilt (*fusarium oxysporum* f. sp. *ciceri*), dry root rot (*Rhizoctonia bataticola*), Ascochyta blight (*Ascochyta rabici*) and collar rot (*Sclerotium rolfsi*). Among all dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler is one of the major soil borne diseases of chickpea.

Typical symptoms of the disease are plants becomes straw colored as dry grass at pod formation and grain filling stage, root becomes black, rotted and brittle shredding of the bark, roots broken easily, minute dark black sclerotial bodies on root surface and in the pith region.

A roving survey for recording chickpea dry root rot disease incidence was undertaken in Marathawada region. The highest disease incidence was observed in Latur district (23 per cent) and lowest in Aurangabad district (10.20 per cent).

The fungus (*R. bataticcola*) was isolated from naturally infected diseased roots on potato dextrose agar. The fungus produced grey to black colored colony. Pathogenicity of the *R. bataticcola* was proved on one month old seedlings of chickpea cv. JG-62.

Morphological studies showed that the mycelium of *R. bataticcola* initially appeared as whitish mycelium but later turned to dark gray to blackish due to sclerotial formation. Mycelium was septate and sclerotia were dark black to dark blackish and irregular to round, oval, spherical in shape and irregular measured diameter 68.19 (48.71 × 87.17) µm in size.

Cultural characters of the fungus were studied on nine solid media. The excellent growth and sporulation was observed on potato dextrose agar and host extract agar medium (chickpea) followed by carrot agar, sabouraud's agar and Richards' agar. The colony of *Rhizoctonia bataticola* was circular, light black with

clear zonation and thick growth on the upper surface of PDA, which turned into blackish on ageing.

In *in vitro* total nine fungicides, were evaluated at different concentrations. The pathogen was most sensitive to carbendazim (100 per cent inhibition) followed by benomyl (93.61 per cent inhibition), carbendazim+mancozeb (92.91 per cent inhibition), mancozeb (91.83 per cent inhibition), thiram (83.66 per cent inhibition), propiconazole (74.58 per cent inhibition), captan (72.08 per cent inhibition) And least inhibition was recorded in carboxin (61.17 per cent) .

The suppression of growth of pathogen was maximum in *Trichoderma viride* (86.62 per cent) followed by *Trichoderma harzianum*(66.33 per cent), *T. hamatum* (61.65 per cent), *T.longibachiatum* (57.75 per cent), *T. koningii* (56.53 per cent), *T. virens* (47.28 per cent)and the least effective bioagent was *Aspergillus niger* (55.13 per cent). *Trichoderma viride* showed best performance against the pathogen.

All of the seven organic amendments evaluated in pot culture (sick soil) as pre-sowing soil application were found effective against *R. bataticola*. However, neem seed cake was found most effective with significantly highest seed germination (83.51%), lowest average mortality (20.05%) and highest average mortality reduction (72.88%). This was followed by the amendments viz., FYM (73.39%, 29.12% and 63.31%), vermicompost (66.71%, 36.59% and 51.23%), mustard cake (60.00%, 42.16% and 43.63%) and castor cake (56.69%, 45.59% and 39.25%), of the seed germination, average mortality, average mortality reduction, respectively. Whereas, poultry manure was found least effective with significantly least seed germination (40.46%), maximum average mortality (47.72%) and minimum average mortality reduction (36.17%).

The three fungicides, three bioagents and two organic amendments were evaluated by seed treatment method in pot culture experiment with chickpea cultivar JG 62 against *R.bataticola* .Observations on seed germination , per cent disease incidence and per cent inhibition over control were recorded. Among three groups in chemical group carbendazim was found best which gave maximum germination count (93.33 per cent) with average dry root rot incidence(31.15 per cent) and maximum disease control (71.09 per cent) as compared to benomyl and carbendazim+mancozeb and control. In biocontrol group *T.viride* was found most effective as compared to other two bioagents (i.e. *T.harzianum*, *T.hamatum*) and *T.viride* gave maximum germination count (90.00 per cent) with less disease

incidence (11.56 per cent) and good disease control (65.91 per cent) over control. In organic amendment neem seed cake was found most effective as compared to FYM. Neem seed cake gave maximum germination count (70.00 per cent) and least disease incidence (50.34 per cent) and good disease control (44.91 per cent).

It is evident from the Table 8 that out of 25 germplasm lines, 09 lines viz. SAKI-9516, BCP-26, BDNG20B-1, JG-315 (R-ch), Rajas, PG-0515, BCP-21, BDNG 13-1, BCP-49 were found resistant. The 06 lines viz PG0302-10, Digvijay, Vishal, JAKI 9218, ICCV-5313, BDNG-9-3 were found moderately resistant. The 01 line i.e. ICCV-5313 was found moderately susceptible. The two lines viz AKG 1103, Vijay were found susceptible. The 07 lines viz BDNG 2013-2, PG9801, JG-62(S-W.Ch), PG0014-6-3, PG-0102-11-2, PG-9758, PG4333 were found highly susceptible.

CONCLUSIONS :

Thus from the results obtained on various aspects during present investigations on dry root rot (*R.bataticola*) of chickpea (*Cicer arietinum*), it can be concluded that:

- *R.bataticola* is one of the most destructive pathogen causing dry root rot diseases in chickpea and there by inflicting accountable yield losses.
- Survey report in Marathawada region showed that the dry root rot of chickpea was severe during *Rabi* season in dry land soil.
- The pathogen (*R.bataticola*). grew and proliferated better on a wide range of agar culture media preferably on., potato dextrose agar.
- The mycelium of *R. bataticola* was septate and sclerotia were dark black to dark blackish and irregular to round, oval, spherical in shape and irregular measured 68.19 μm (48.71×87.17) in diameter size.
- Studies (*in vitro*) on efficacy of fungicides, bioagents and organic amendments against *R.bataticola* showed that, carbendazim, benomyl and carbendazim+mancoze were effective fungicides to control the dry root rot of chickpea. Among the bioagents *T. viride* was best to control dry root rot followed by *T. harzianum* and in organic amendments neem seed cake and FYM could effectively and economically manage/control dry root rot of chickpea.
- In pot culture studies, among three groups in chemical group carbendazim was found best which gave maximum germination count (93.33 per cent) with average dry root rot incidence of 31.15 per cent In biocontrol group, *T.viride*

was found most effective. In organic amendments, neem seed cake was found most effective.

- In field screening out of 25 germplasm line 09 lines were found resistant. Six lines were found moderately resistant. One line was found moderately susceptible. Two lines were found susceptible. Seven lines were found highly susceptible.



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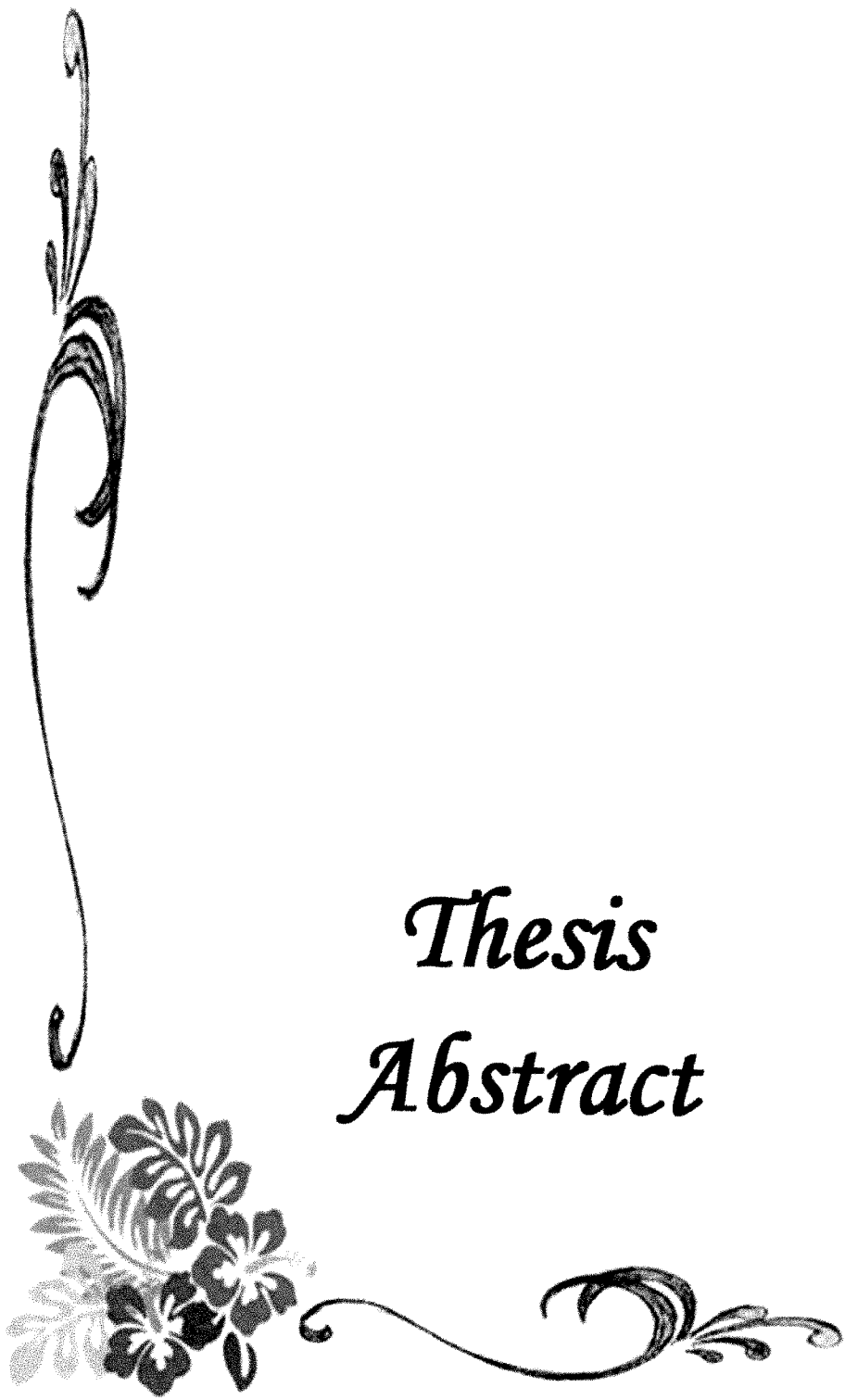
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Thesis
Abstract

“Studies on dry root rot of chickpea incited by *Rhizoctonia bataticola* (Taub) Butler.”

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Abstract

Chickpea (*Cicer arietinum* L) is an important pulse crop of India popularly known as ‘Gram’. In India, it is cultivated on nearly 10.22 million ha. area with an annual production of 09.88 metric tonnes and productivity of 967 kg/ha. The Marathwada region contributes 6.58 lakh ha. area with production of 06.64 lakh tonnes and productivity of 918 kg/ha., during *Rabi* (Anonymous, 2014). Of the major fungal diseases infecting chickpea, dry root rot incited by *Rhizoctonia bataticola* (taub) Butler is one of the most destructive and wide spread disease which cause average yield losses of 05-50% (Anonymous, 2012). Keeping in view, economic importance of chickpea and losses incurred by dry root rot disease, present investigations on the aspects viz., survey, isolation, identification and pathogenicity, effect of culture media, *in vitro* evaluation of fungicides and bioagents, *in vitro* (pot culture) evaluation organic amendments, integrated disease management, and germplasm screening against *R. bataticola* were undertaken during *Rabi*, 2014-15 at the Department of Plant Pathology, College of Agriculture, Badnapur.

A roving survey for recording chickpea dry root rot disease incidence was undertaken in Marathawada region. The highest disease incidence was observed in Latur district (23 per cent) and lowest in Aurangabad district (10.20 per cent).

The fungus (*R. bataticcola*) was isolated from naturally infected diseased roots on potato dextrose agar medium. The pathogenicity of *R. bataticcola* was proved on one month old seedlings of chickpea cv JG-62. Morphological studies showed that the mycelium of *R. bataticcola* initially appeared as whitish mycelium but later turned to dark gray to blackish due to sclerotial formation. Mycelium was septate and sclerotia were dark black to dark blackish and irregular to round, oval, spherical in shape and irregular measured 68.19 μ m (48.71 $\times$ 87.17) in diameter size.

Cultural characters of fungus were studied on nine solid media. Excellent growth and sporulation was observed on potato dextrose agar and host extract agar medium (chickpea) followed by carrot agar, Saboround’s agar and

Richards' agar. The colony of *Rhizoctonia bataticola* was circular, light black with clear zonation and thick growth on the upper surface of PDA, which turned into blackish on ageing.

In *in vitro* total nine fungicides, were evaluated at different concentrations. The pathogen was most sensitive to carbendazim (100 per cent inhibition) followed by benomyl (93.61 per cent inhibition) and carbendazim+mancozeb (92.91 per cent inhibition). Least inhibition was recorded in carboxin (61.17 per cent).

Suppression of growth of the pathogen was maximum in *Trichoderma viride* (86.62 per cent) followed by *Trichoderma harzianum* (66.33 per cent) and the least effective bioagent was *Aspergillus niger* (55.13 per cent). *Trichoderma viride* showed best performance against the pathogen.

All of the seven organic amendments evaluated in pot culture (sick soil) as pre-sowing soil application were found effective against *R. bataticola*. However, neem seed cake was found most effective with significantly highest seed germination (83.51%), lowest average mortality (20.05%) and highest average mortality reduction (72.88%). This was followed by the amendments viz., FYM (73.39%, 29.12% and 63.31%) and vermicompost (66.71%, 36.59% and 51.23%) for seed germination, average mortality and average mortality reduction.

The three fungicides, three bioagents and two organic amendments were evaluated by seed treatment method in pot culture experiment against *R. bataticola*. Among three groups in chemical group carbendazim was found best which gave maximum germination count (93.33 per cent) with average dry root rot incidence (31.15 per cent) and maximum disease control (71.09 per cent) as compared to benomyl and carbendazim+mancozeb over control. In bio control group *T. viride* was found most effective as compared to other two bioagents i.e. *T. harzianum*, *T. hamatum* and *T. viride* gave maximum germination count (90.00 per cent) less disease incidence (11.56 per cent) and good disease control (65.91 per cent) as compared to control. In organic amendment Neem seed cake was found most effective as compared to FYM. neem seed cake gave maximum germination count (70.00 per cent) and least disease incidence (50.34 per cent) and good disease control (44.91 per cent).

It is evident from the Table 8 that out of 25 germplasm lines, 09 lines viz. SAKI-9516, BCP-26, BDNG20B-1, JG-315 (R-ch), Rajas, PG-0515, BCP-

21,BDNG 13-1,BCP-49 were found resistant. The 06 line viz PG0302-10, Digvijay, Vishal, JAKI 9218, ICCV-5313, BDNG-9-3 were found moderately resistant. The 01 line i.e. ICCV-5313 were found moderately susceptible. The two line viz AKG 1103 and Vijay were found susceptible. The 07 lines viz BDNG 2013-2, PG9801, JG-62(S-W.Ch), PG0014-6-3, PG-0102-11-2,PG-9758, PG4333 were found highly susceptible.



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MASTER OF SCIENCE

In

PLANT PATHOLOGY

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APPENDIX-I

1. COMPOSITION OF CULTURE MEDIA

1.1 Potato Dextrose Agar (1 lit.)

Peeled Potatoes : 200 g
Dextrose : 20 g
Agar Agar : 20g
Distilled water: 1000ml

1.2 Sand maize flour medium (1 lit.)

Maize flour : 10 g
Sand : 90 g
Distilled water: 1000ml

2.0 CONCENTRATION USED IN FORMULATION

1	Per cent	= 10,000 ppm (parts per million)	
		= 10 gms/ liter	
		= 10 gms/ kg	
0.1	Per cent	= 1000 ppm	= 1000 mg/liter
0.01	Per cent	=100 ppm	= 100 mg/liter
0.001	Per cent	=10 ppm	= 10 mg/liter
0.0001	Per cent	= 1 ppm	= 1 mg/liter